

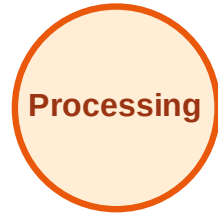
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



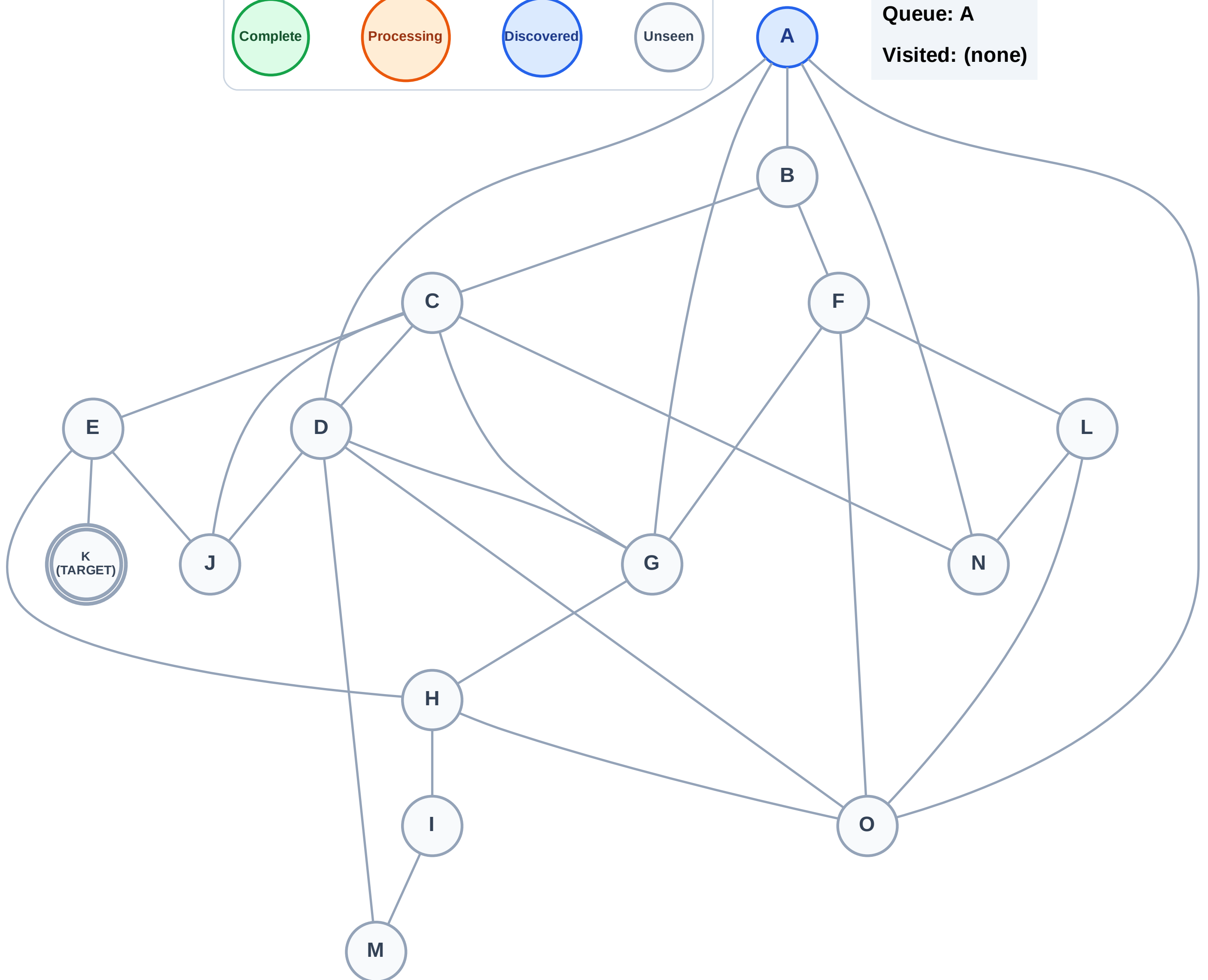
Discovered



Unseen

Queue: A

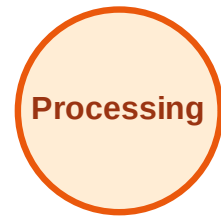
Visited: (none)



BFS Traversal

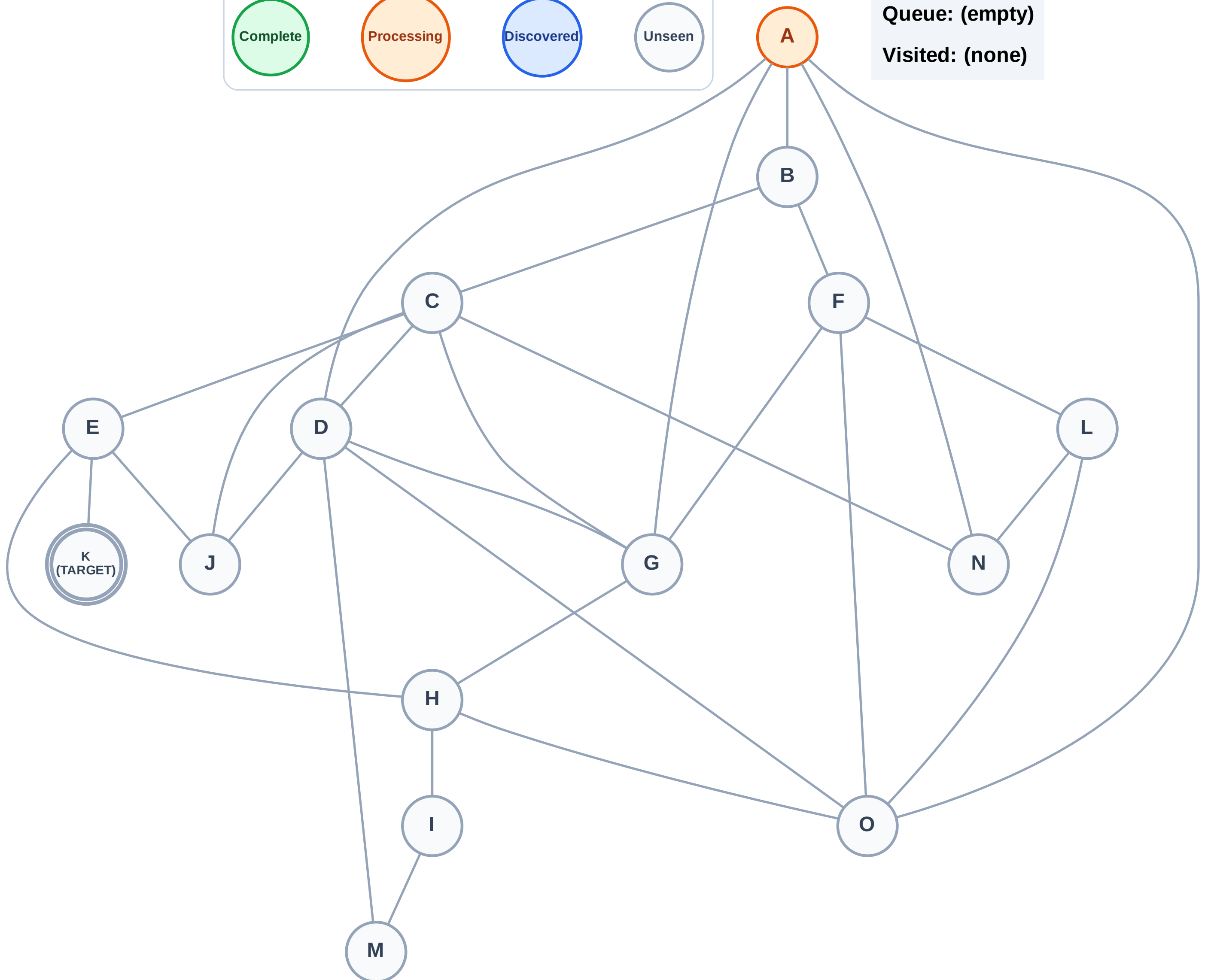
Step 2: Process node A

Legend



Queue: (empty)

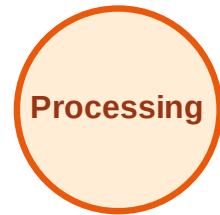
Visited: (none)



BFS Traversal

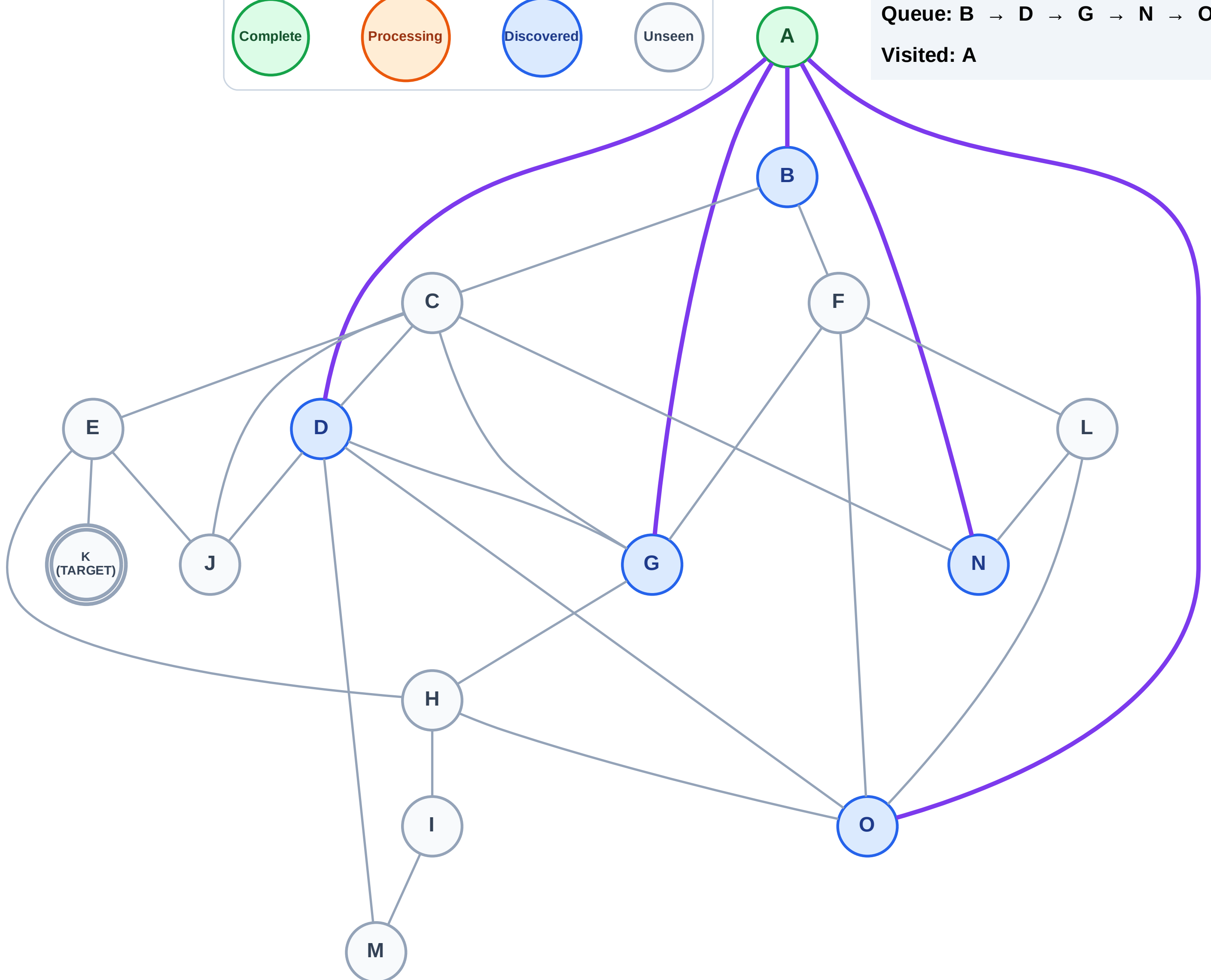
Step 3: Discover B, D, G, N, O from A

Legend



Queue: B → D → G → N → O

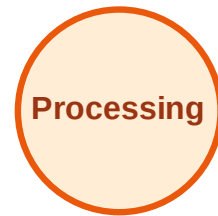
Visited: A



BFS Traversal

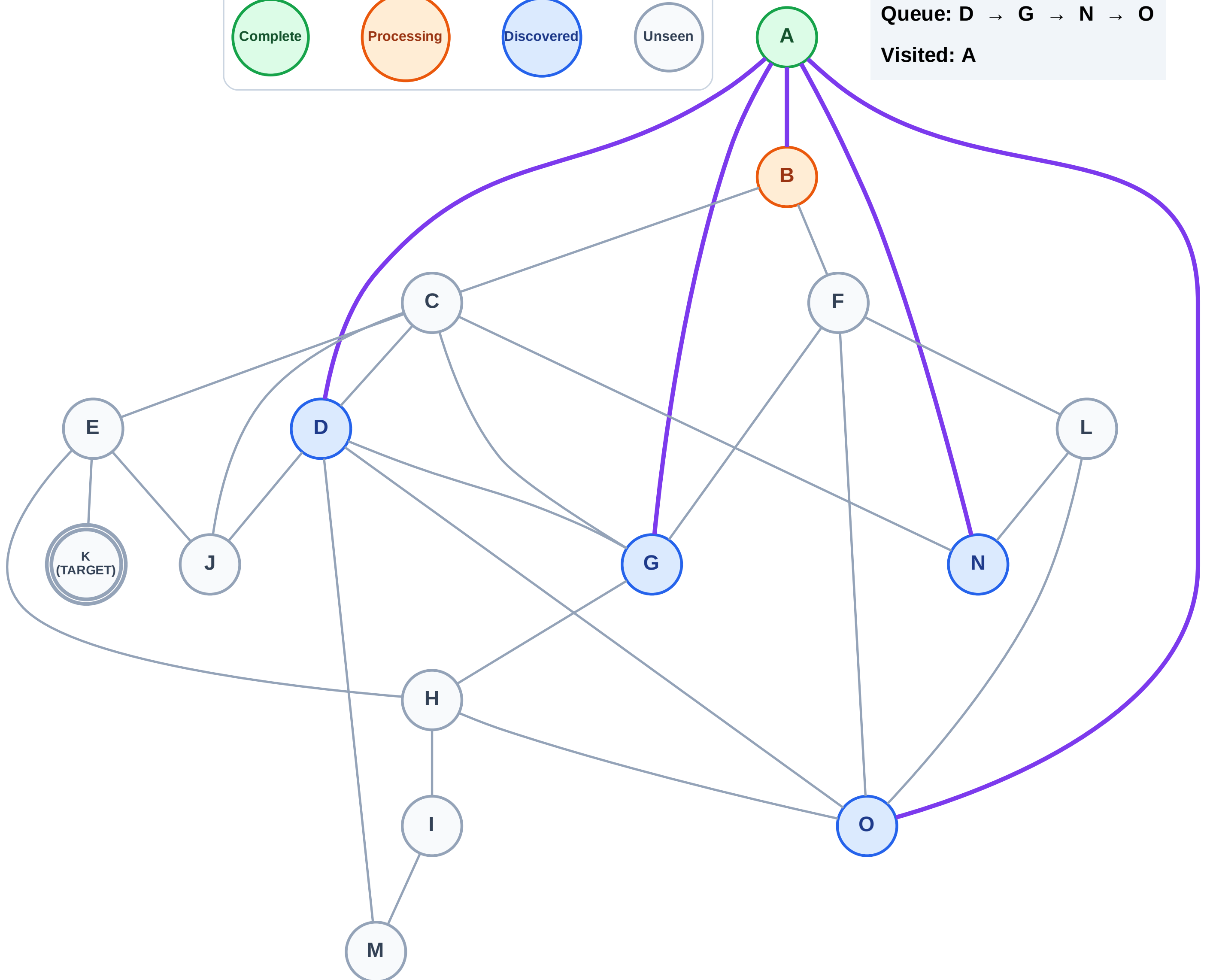
Step 4: Process node B

Legend



Queue: D → G → N → O

Visited: A



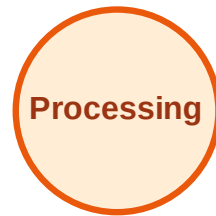
BFS Traversal

Step 5: Discover C, F from B

Legend



Complete



Processing



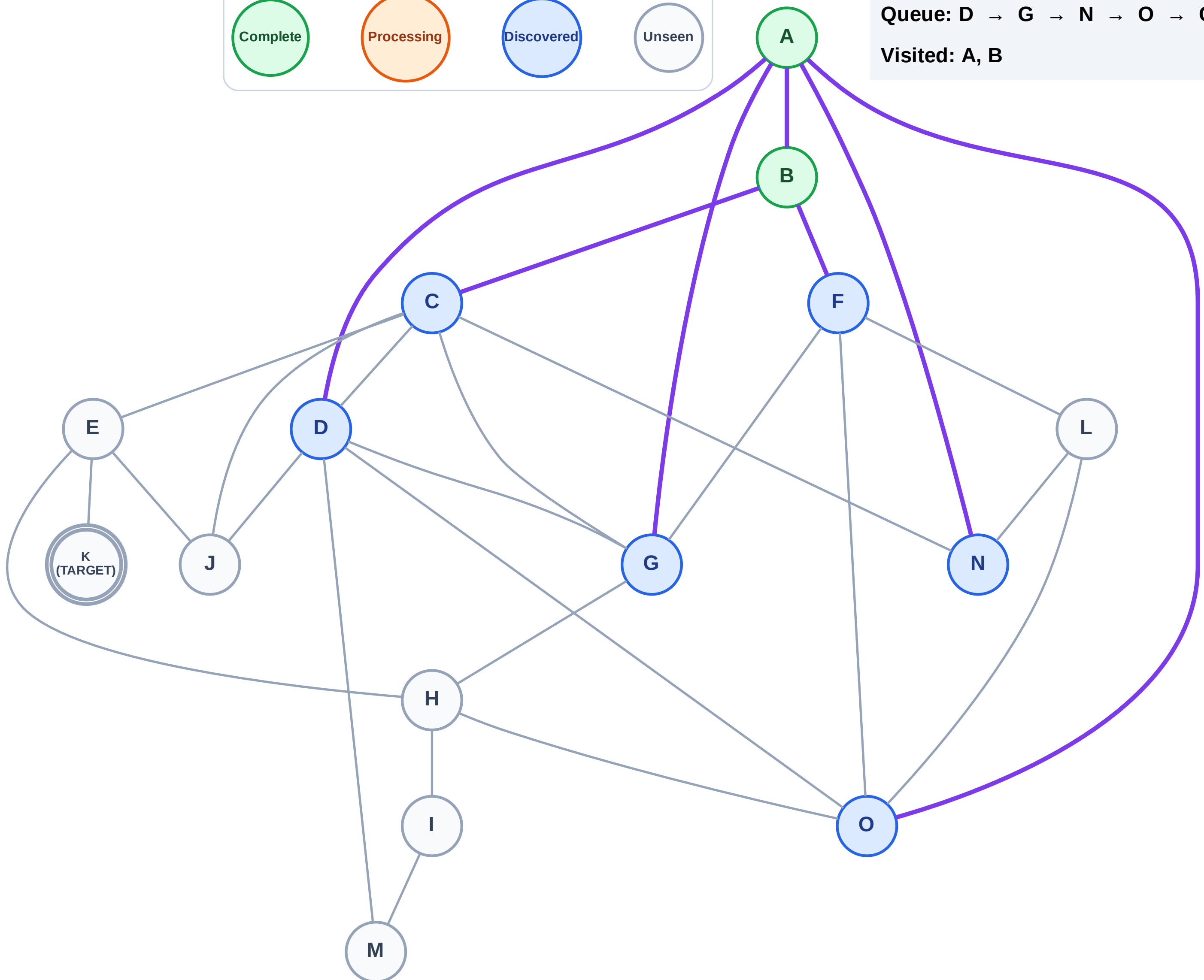
Discovered



Unseen

Queue: D → G → N → O → C → F

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

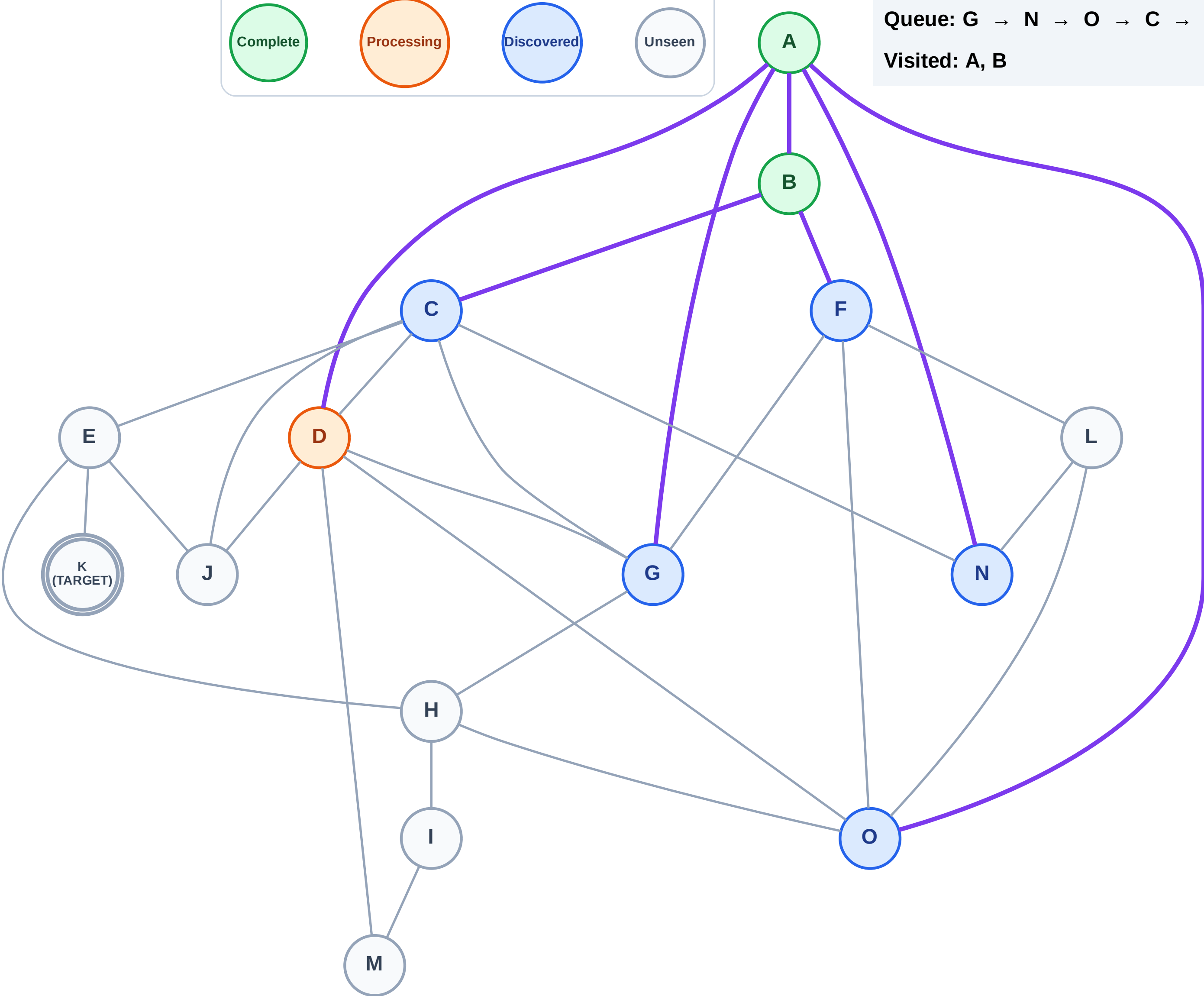
Processing

Discovered

Unseen

Queue: G → N → O → C → F

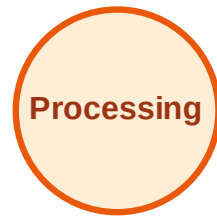
Visited: A, B



BFS Traversal

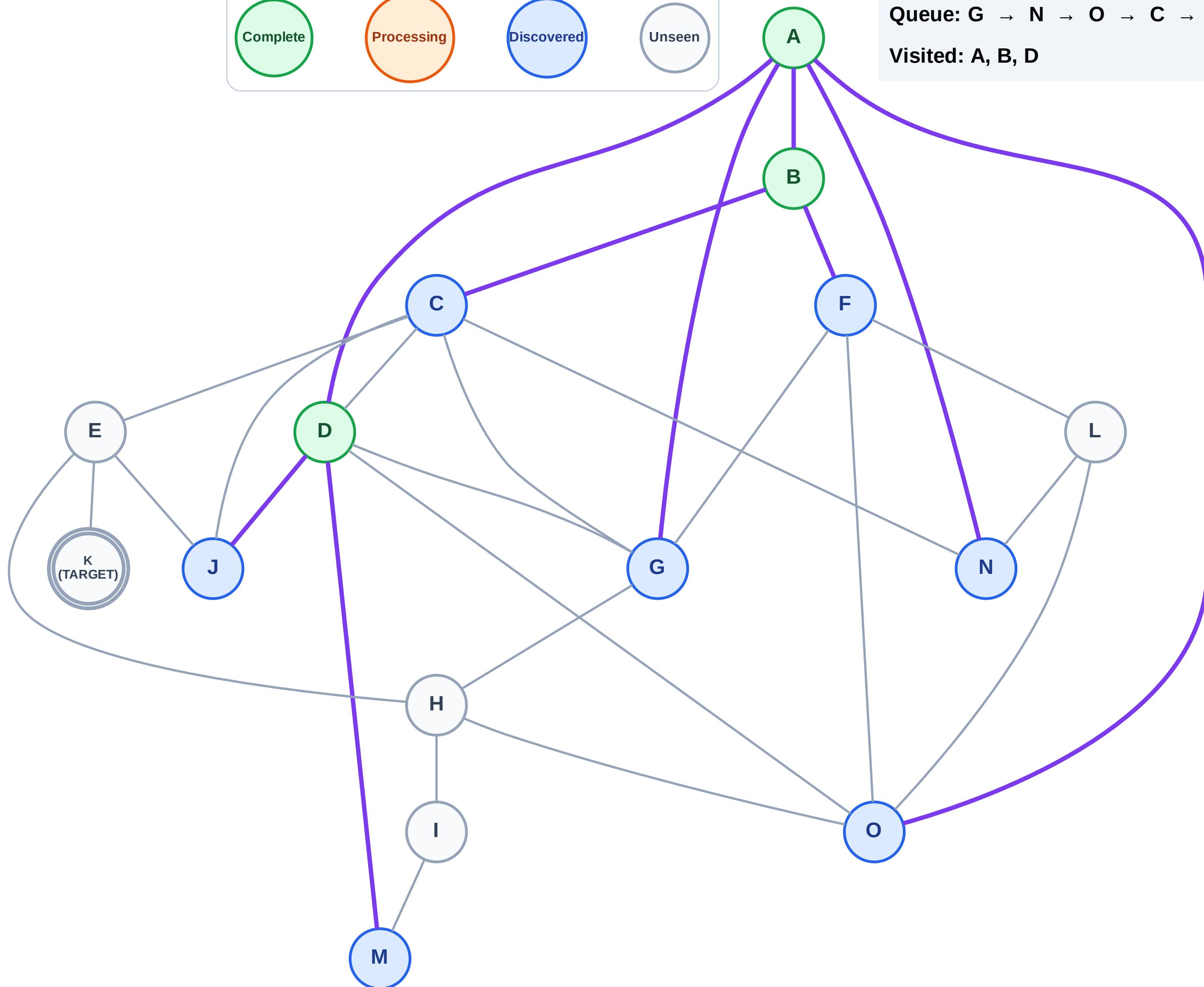
Step 7: Discover J, M from D

Legend



Queue: G → N → O → C → F → J → M

Visited: A, B, D



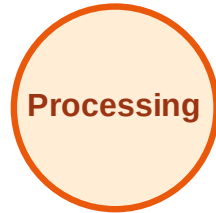
BFS Traversal

Step 8: Process node G

Legend



Complete



Processing



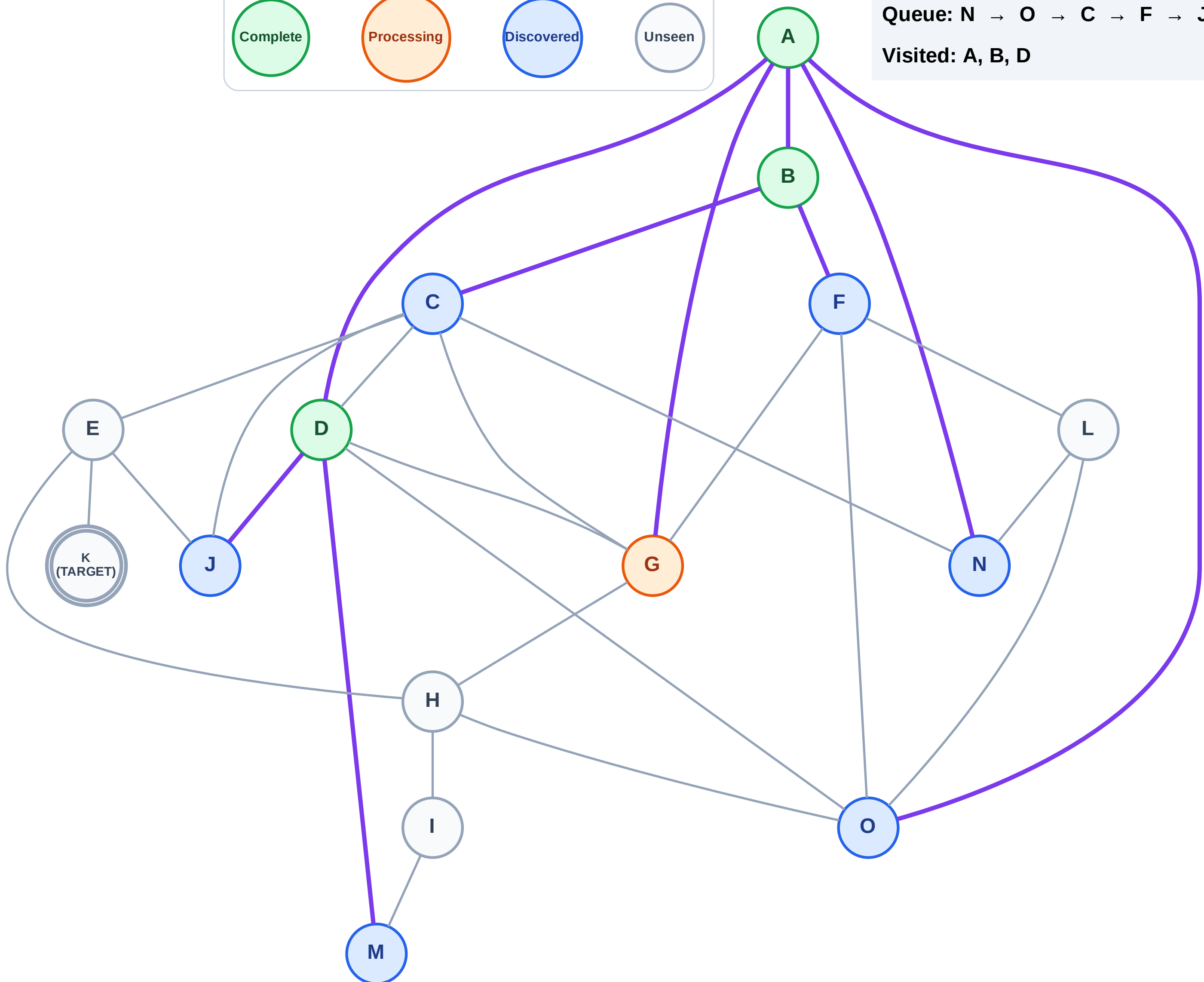
Discovered



Unseen

Queue: N → O → C → F → J → M

Visited: A, B, D



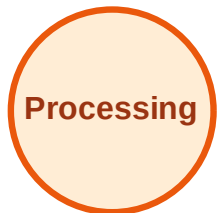
BFS Traversal

Step 9: Discover H from G

Legend



Complete



Processing



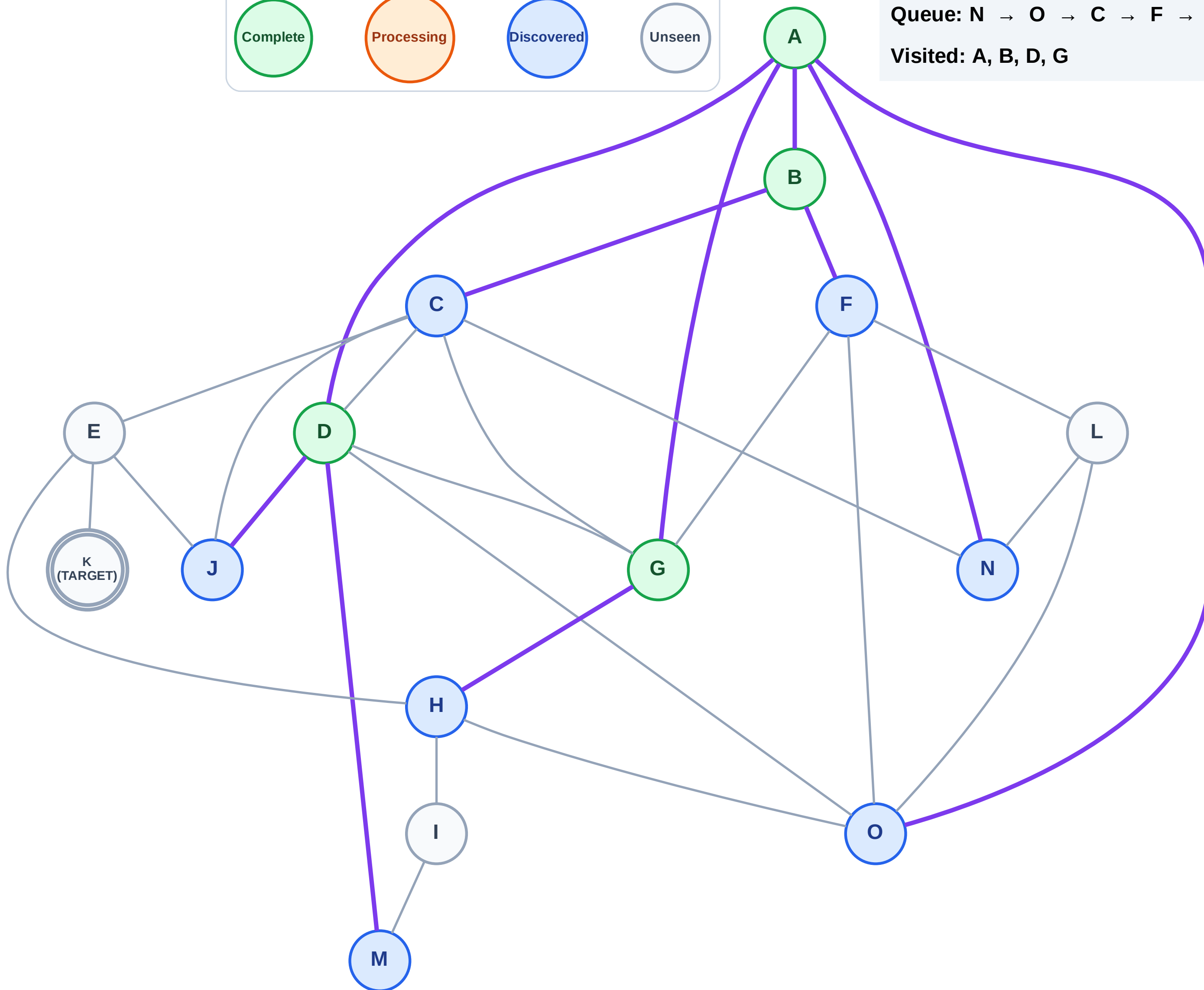
Discovered



Unseen

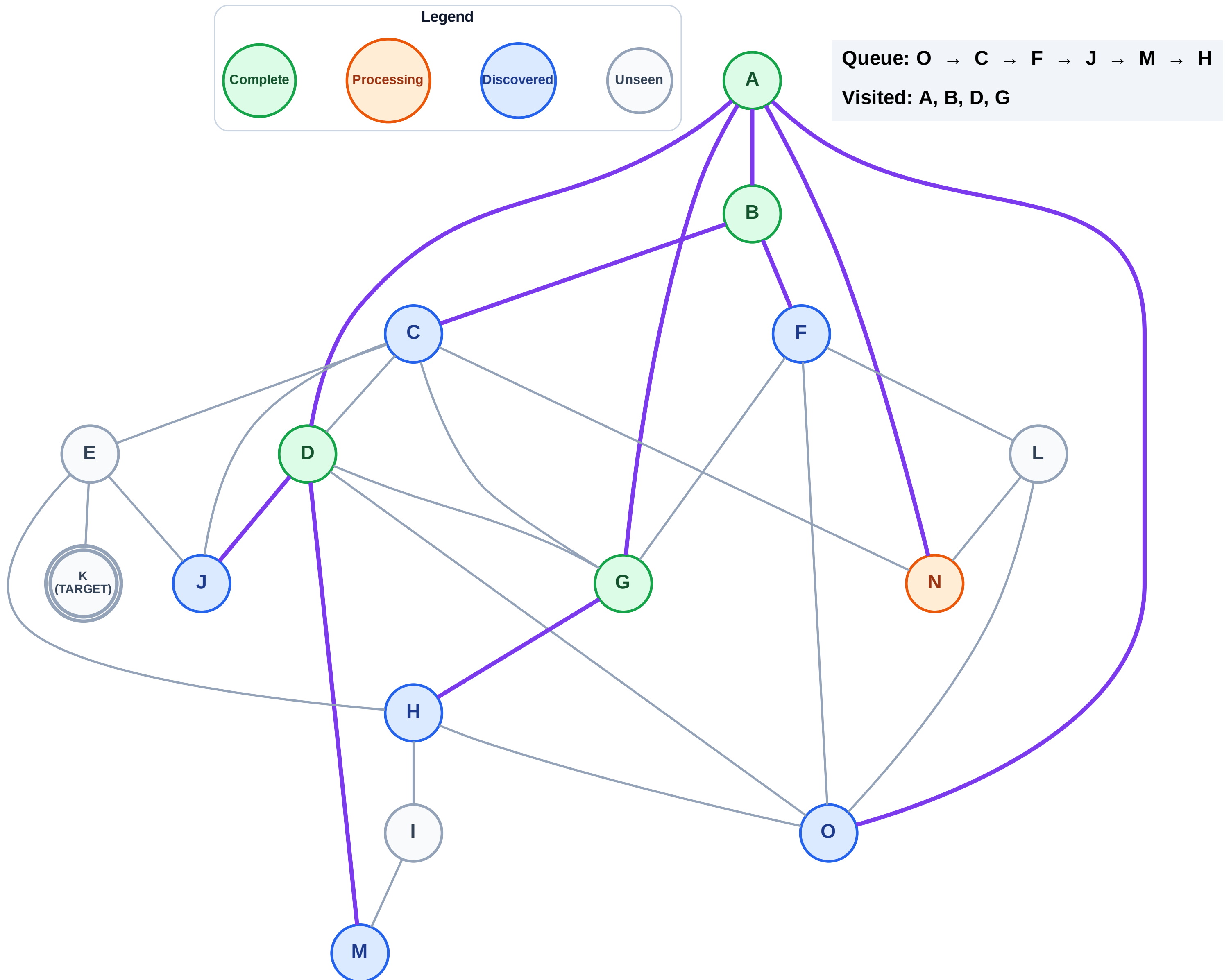
Queue: N → O → C → F → J → M → H

Visited: A, B, D, G



Step 10: Process node N

Step 10: Process node N



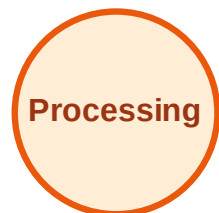
BFS Traversal

Step 11: Discover L from N

Legend



Complete



Processing



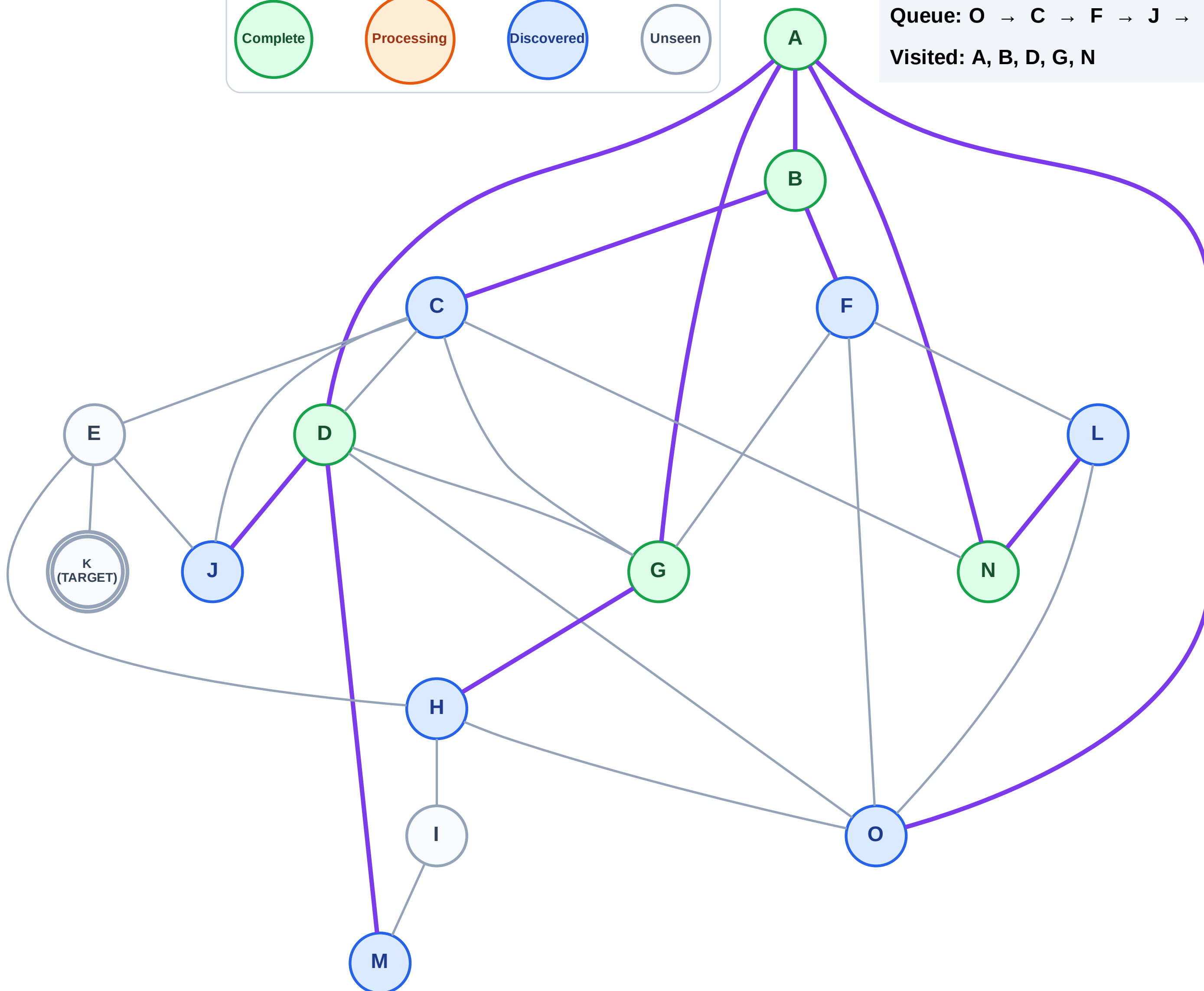
Discovered



Unseen

Queue: O → C → F → J → M → H → L

Visited: A, B, D, G, N



BFS Traversal

Step 12: Process node O

Legend

Complete

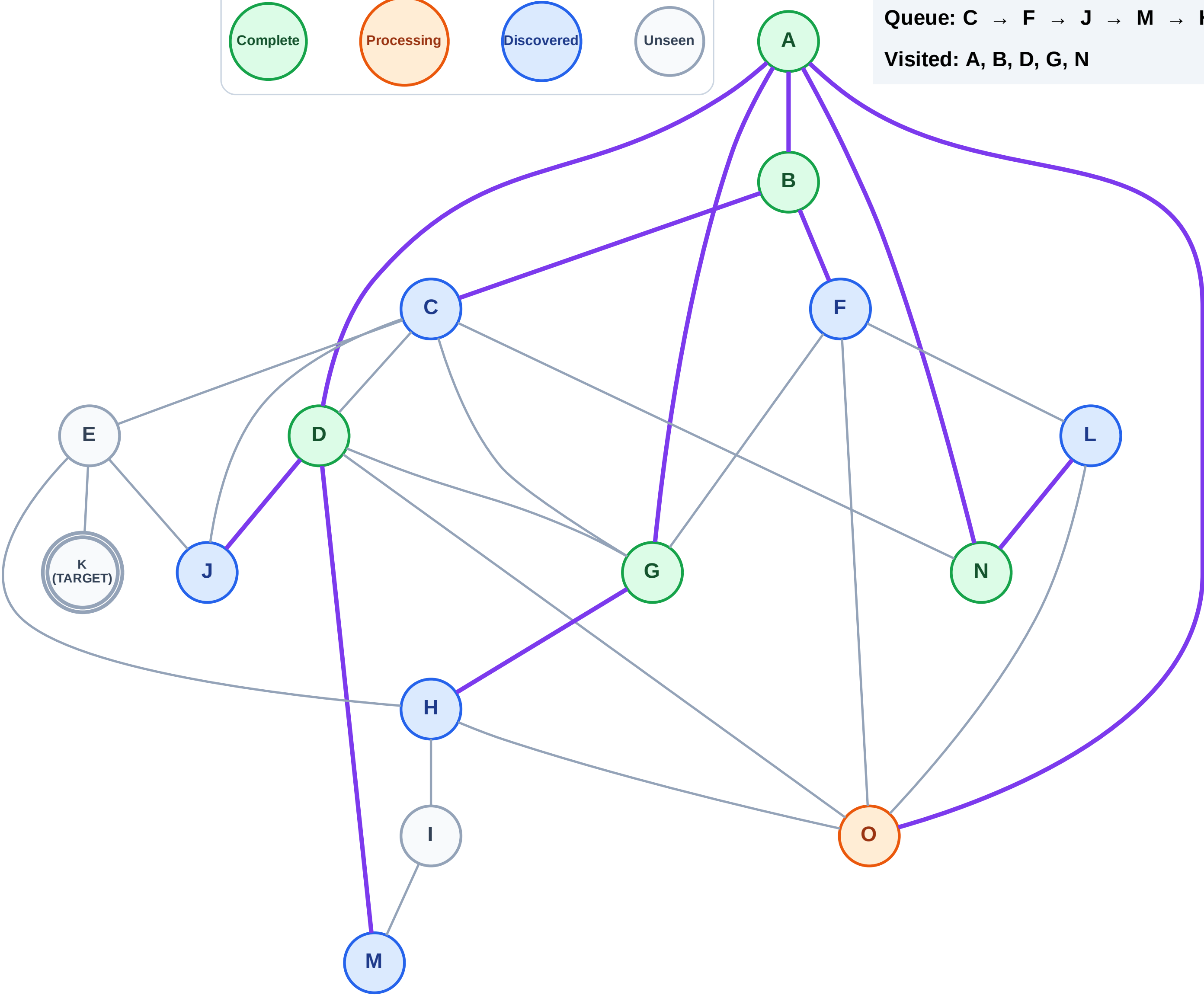
Processing

Discovered

Unseen

Queue: C → F → J → M → H → L

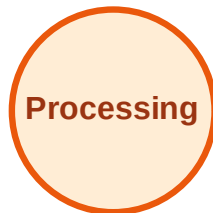
Visited: A, B, D, G, N



BFS Traversal

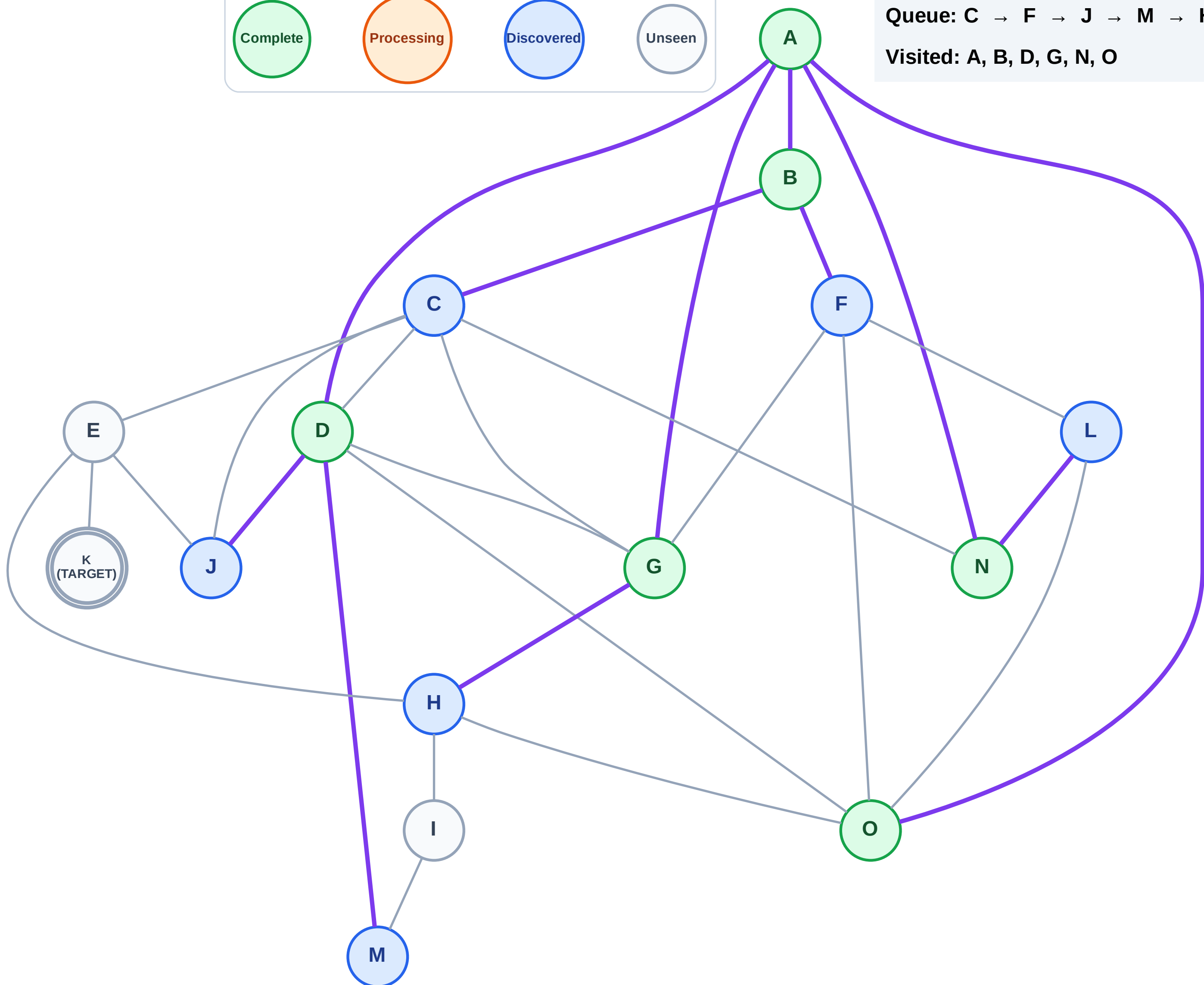
Step 13: No new neighbors from O

Legend



Queue: C → F → J → M → H → L

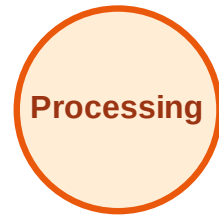
Visited: A, B, D, G, N, O



BFS Traversal

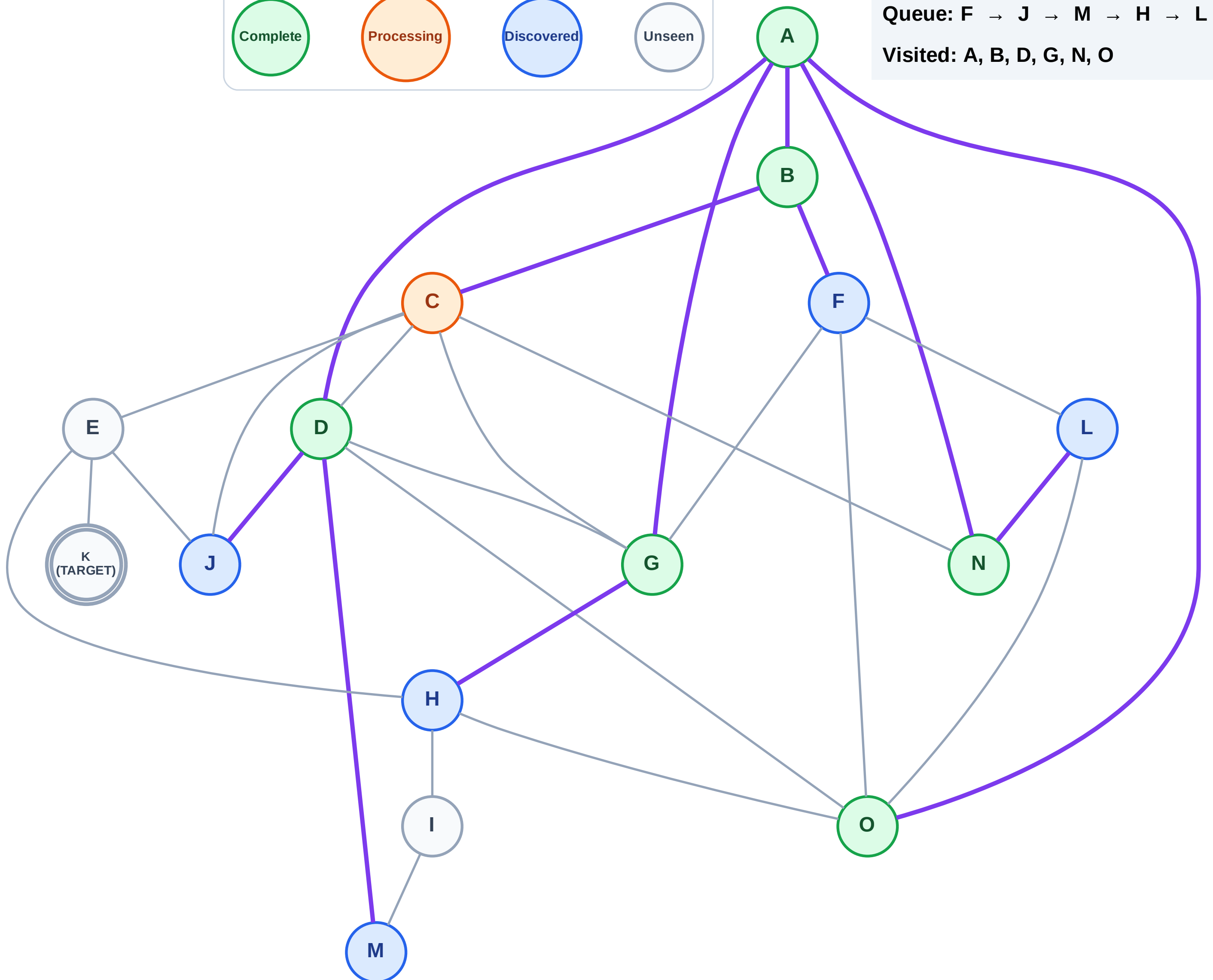
Step 14: Process node C

Legend



Queue: F → J → M → H → L

Visited: A, B, D, G, N, O



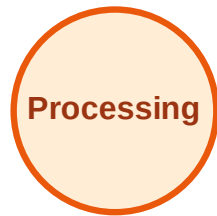
BFS Traversal

Step 15: Discover E from C

Legend



Complete



Processing



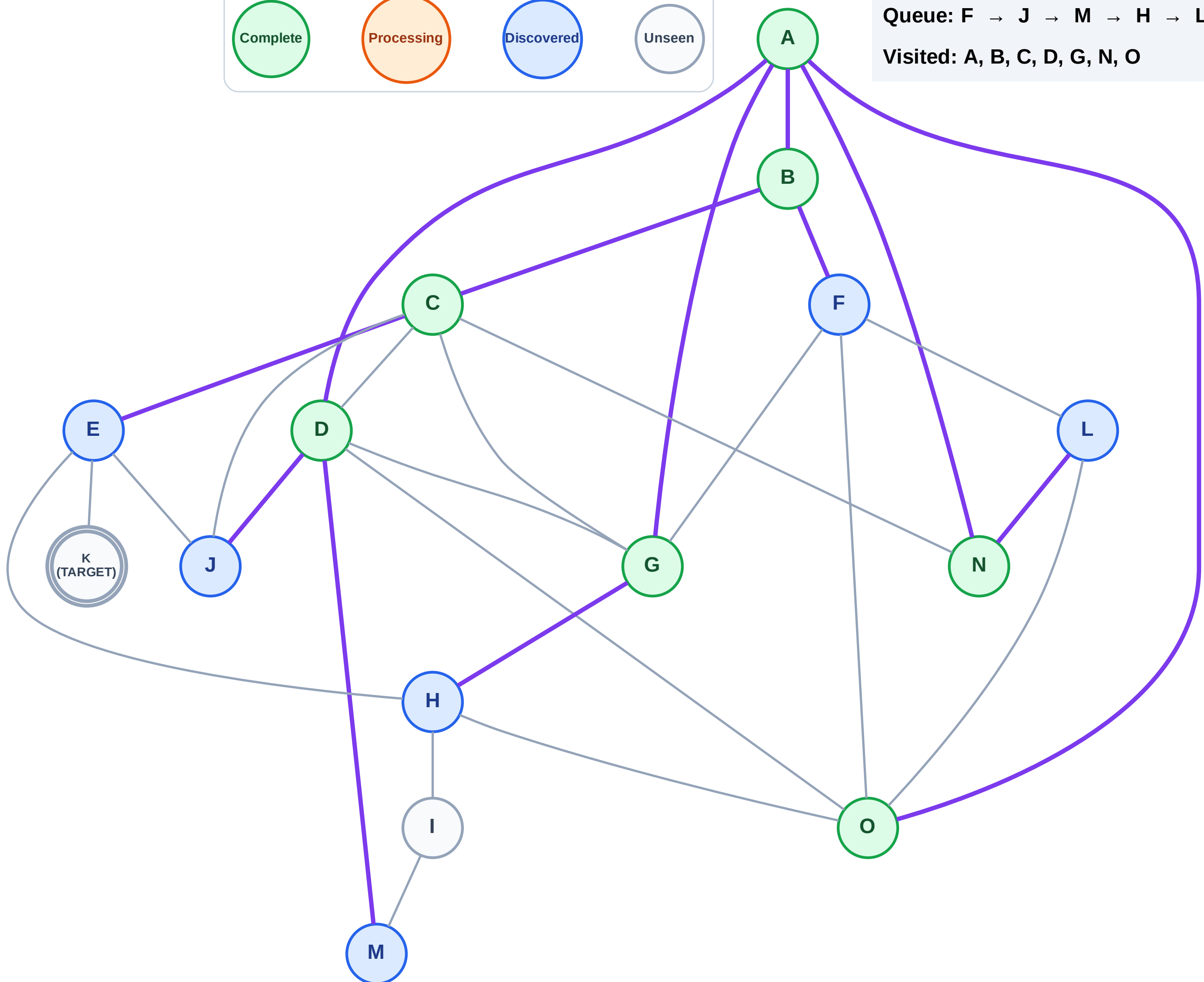
Discovered



Unseen

Queue: F → J → M → H → L → E

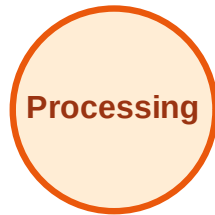
Visited: A, B, C, D, G, N, O



BFS Traversal

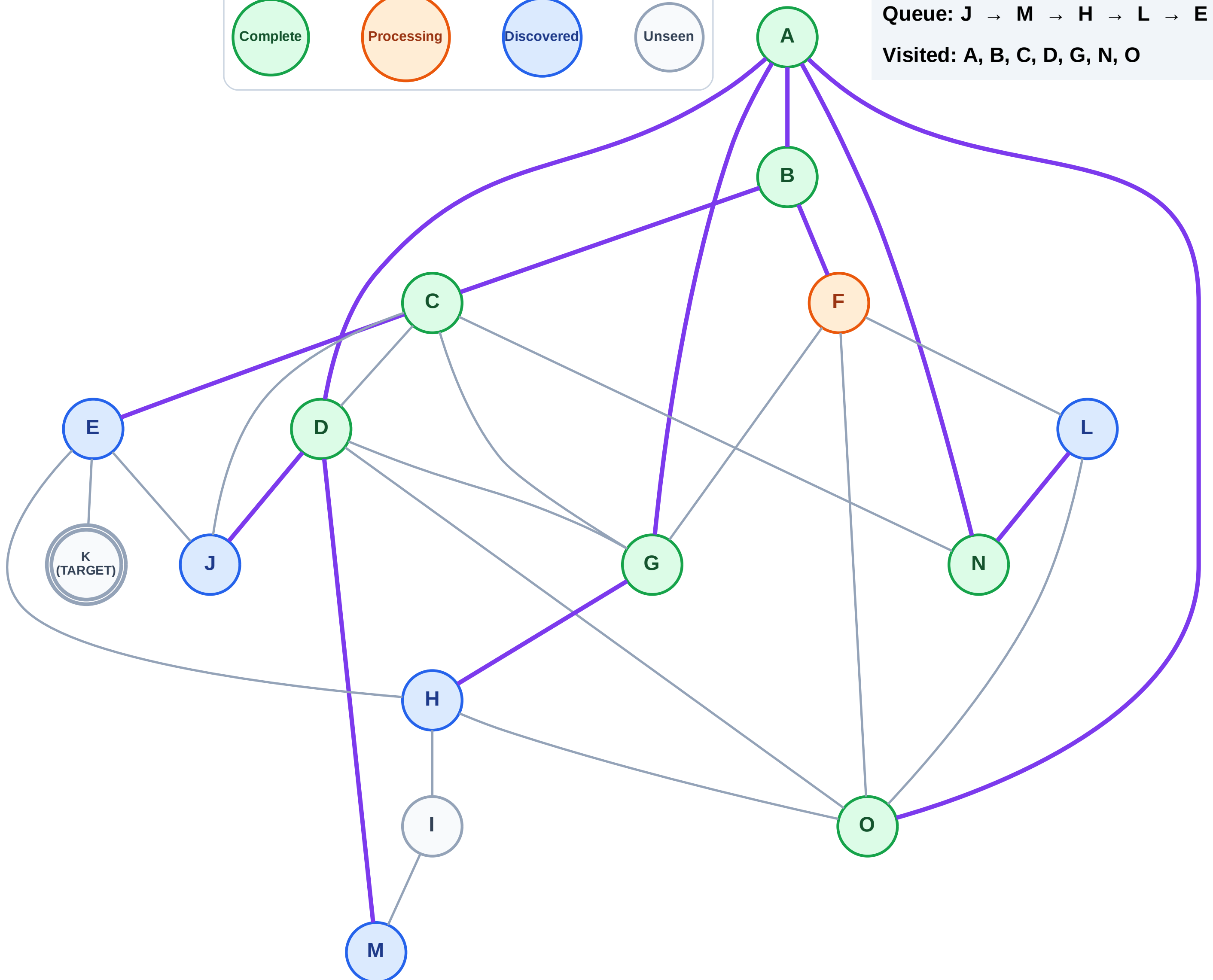
Step 16: Process node F

Legend



Queue: J → M → H → L → E

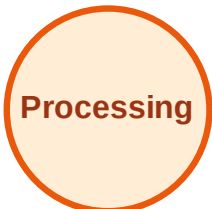
Visited: A, B, C, D, G, N, O



BFS Traversal

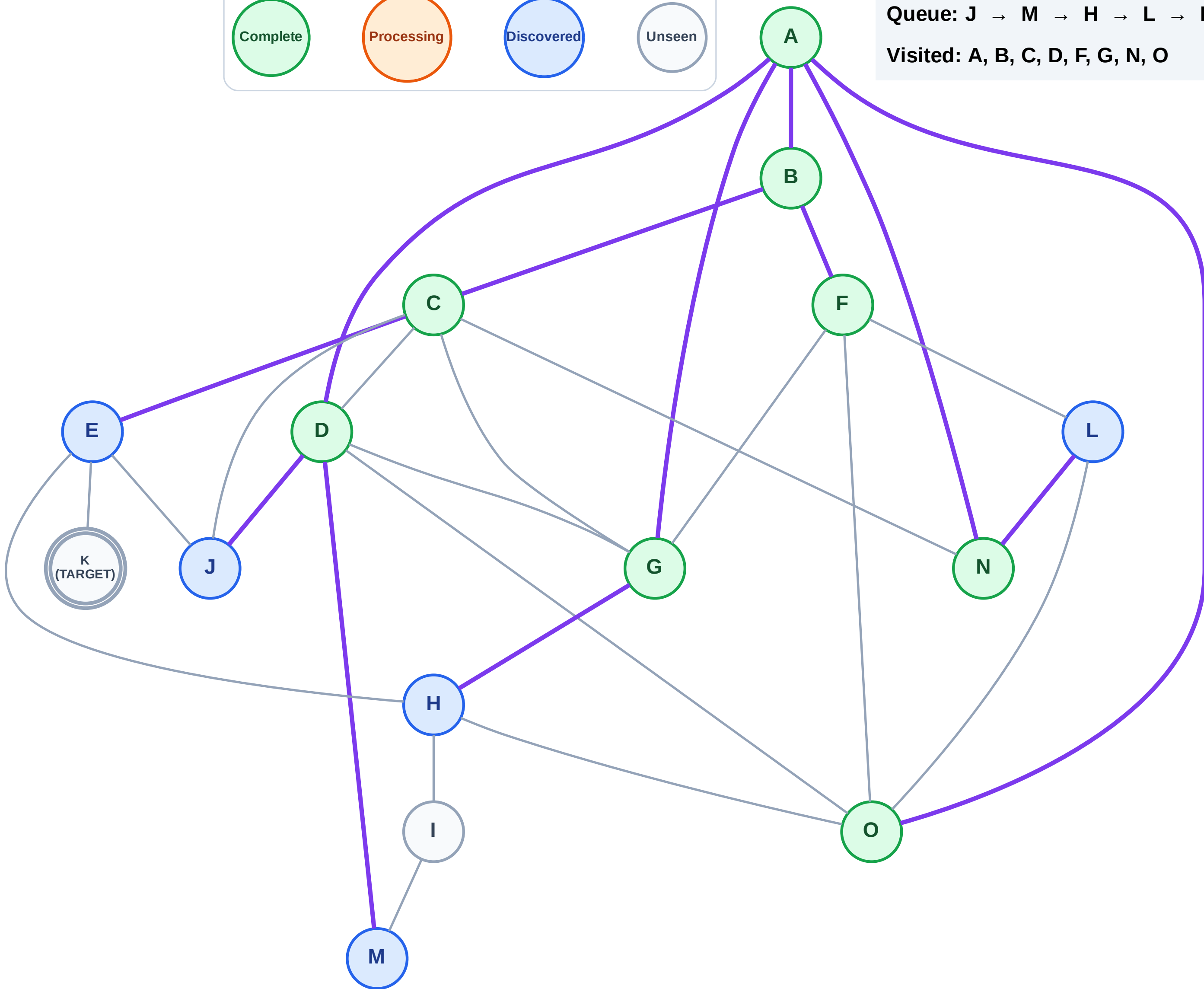
Step 17: No new neighbors from F

Legend



Queue: J → M → H → L → E

Visited: A, B, C, D, F, G, N, O



BFS Traversal

Step 18: Process node J

Legend

Complete

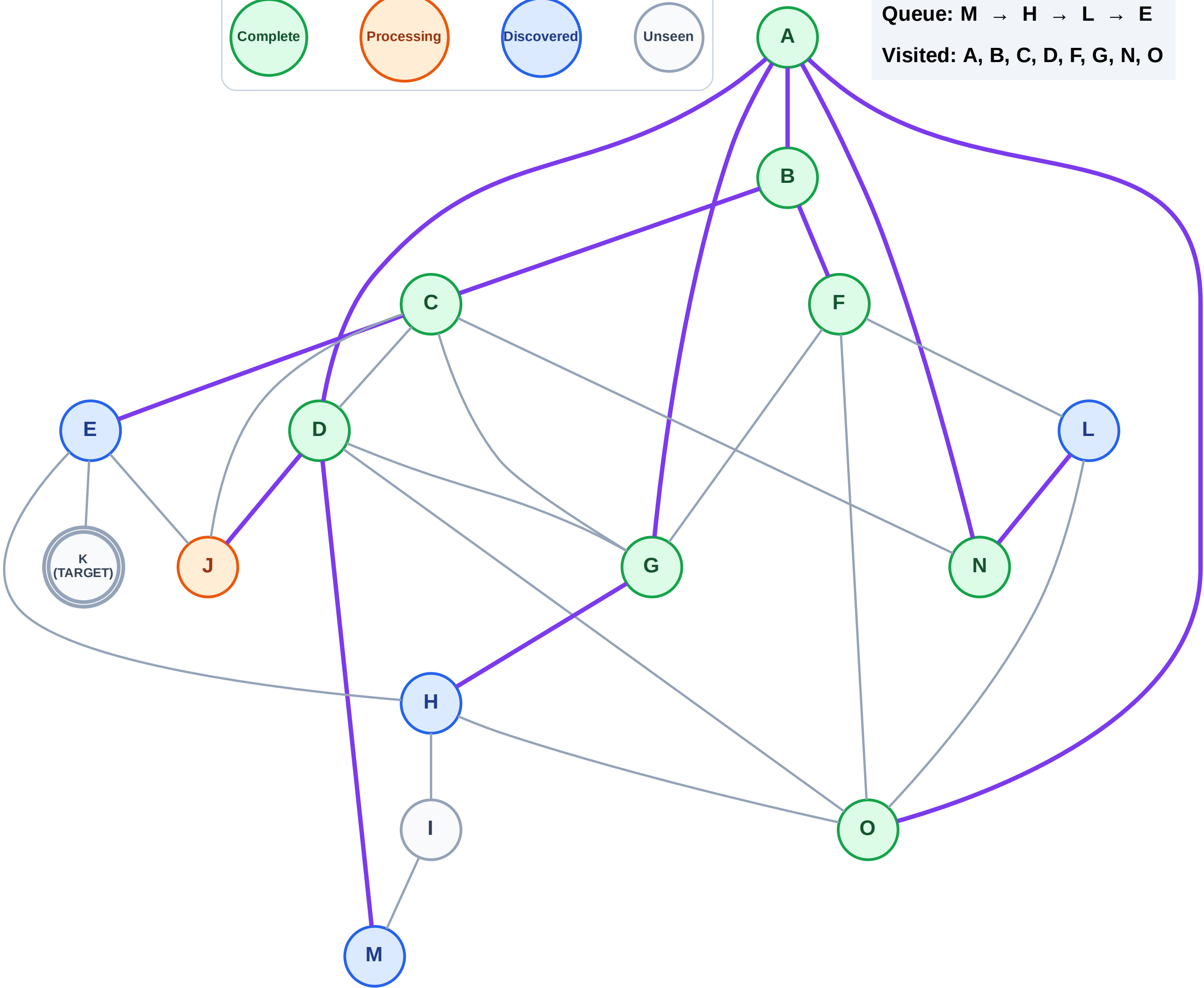
Processing

Discovered

Unseen

Queue: M → H → L → E

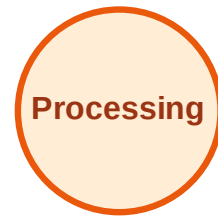
Visited: A, B, C, D, F, G, N, O



BFS Traversal

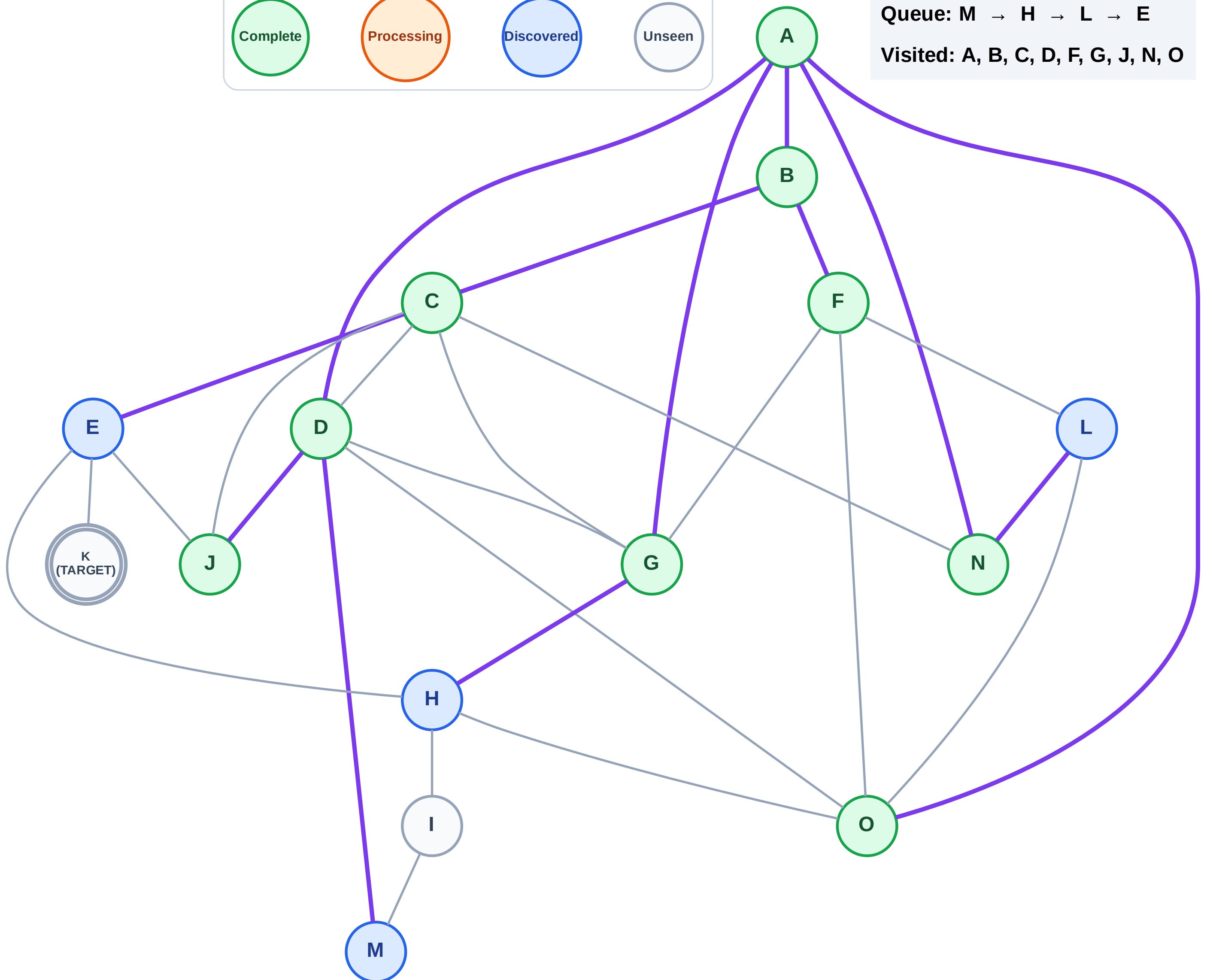
Step 19: No new neighbors from J

Legend



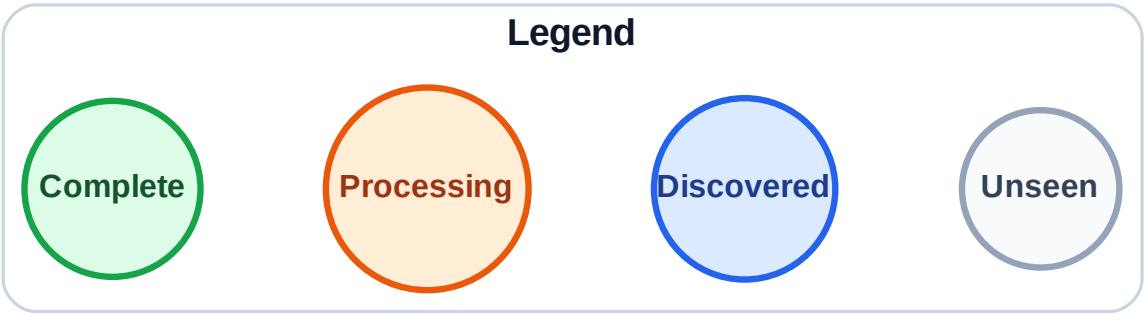
Queue: M → H → L → E

Visited: A, B, C, D, F, G, J, N, O



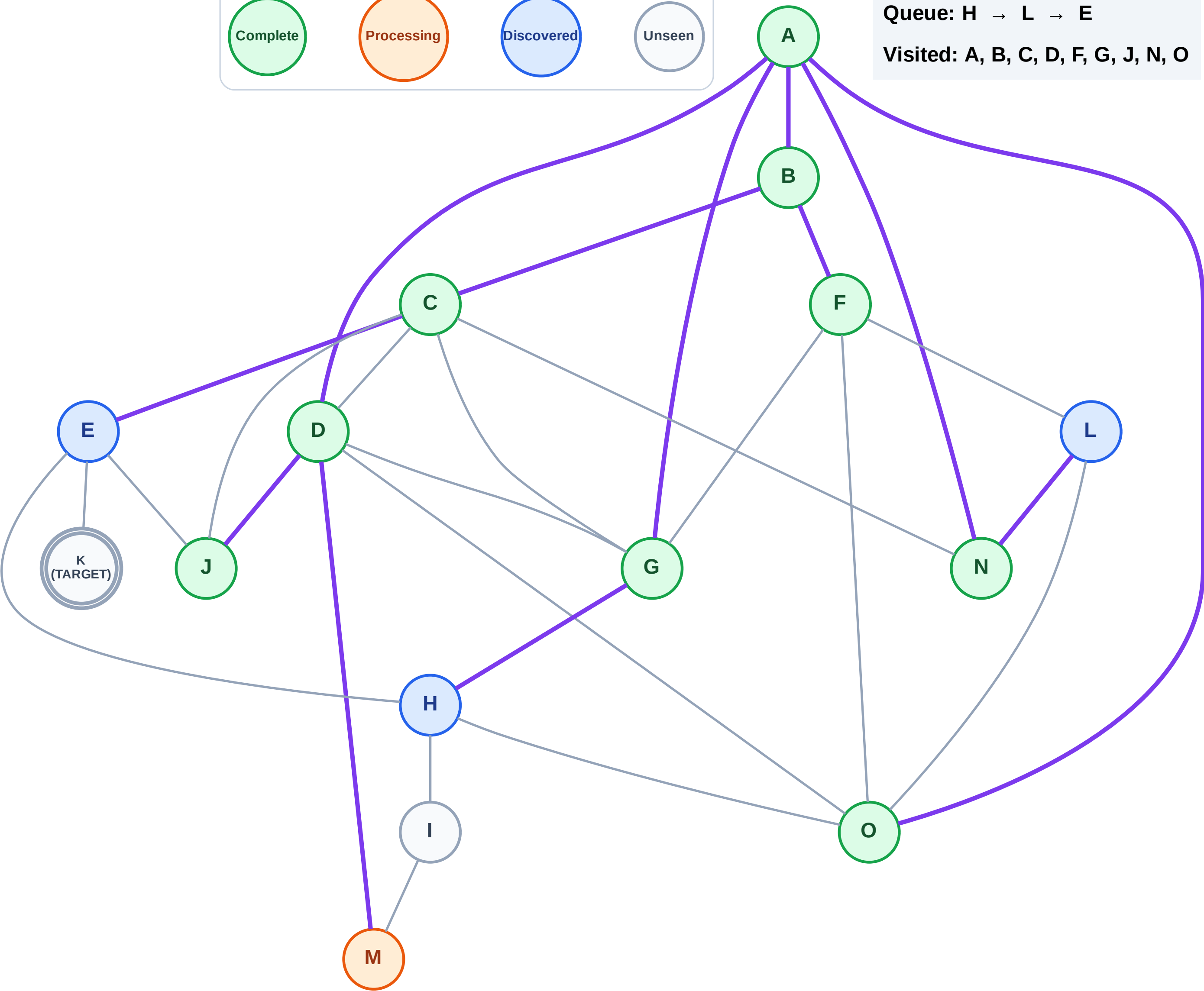
BFS Traversal

Step 20: Process node M



Queue: H → L → E

Visited: A, B, C, D, F, G, J, N, O



Step 21: Discover I from M

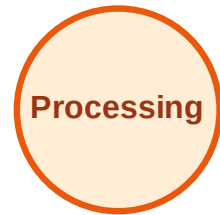
Step 21: Discover I from M



BFS Traversal

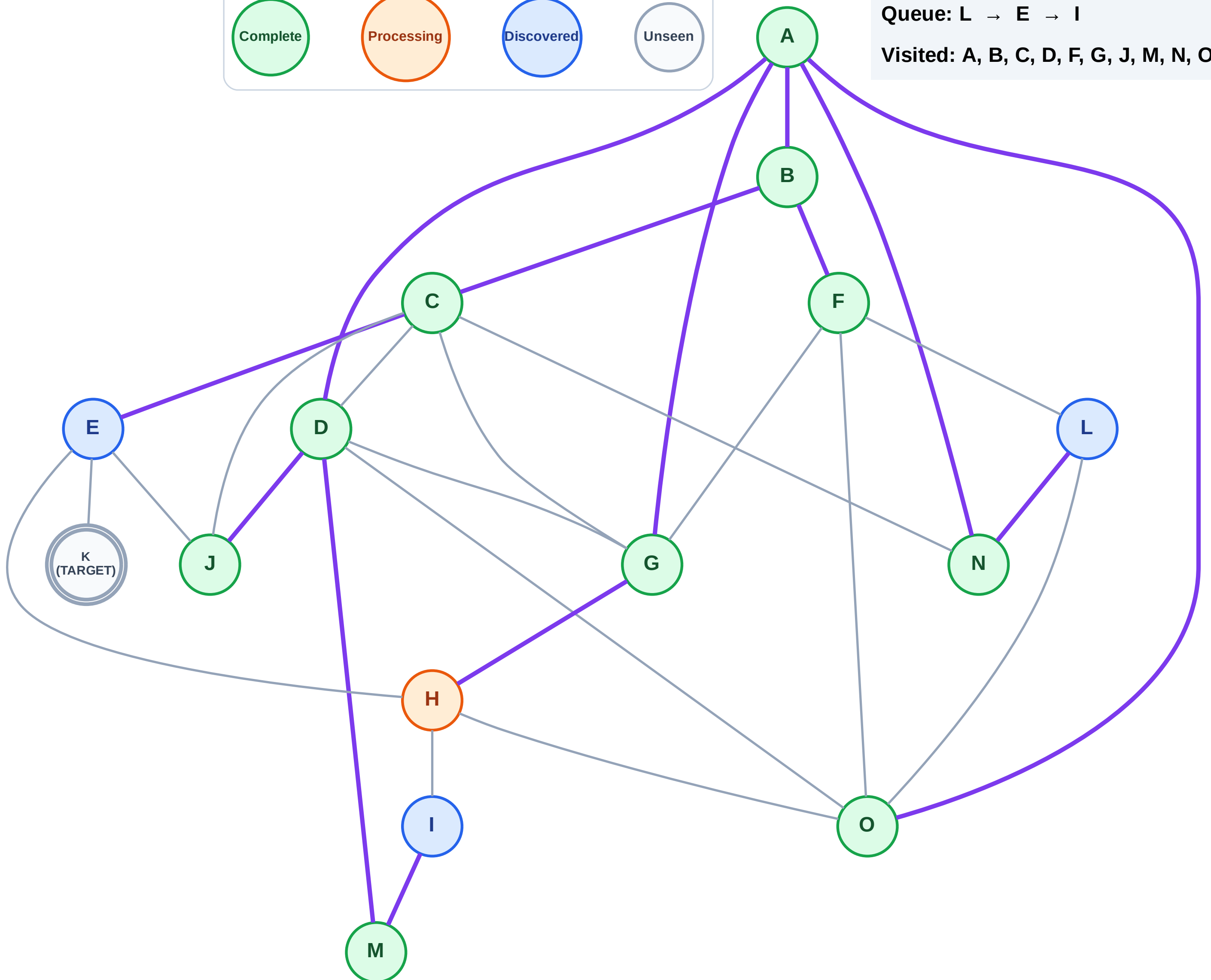
Step 22: Process node H

Legend



Queue: L → E → I

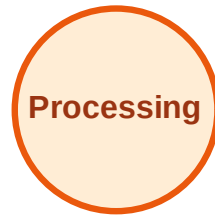
Visited: A, B, C, D, F, G, J, M, N, O



BFS Traversal

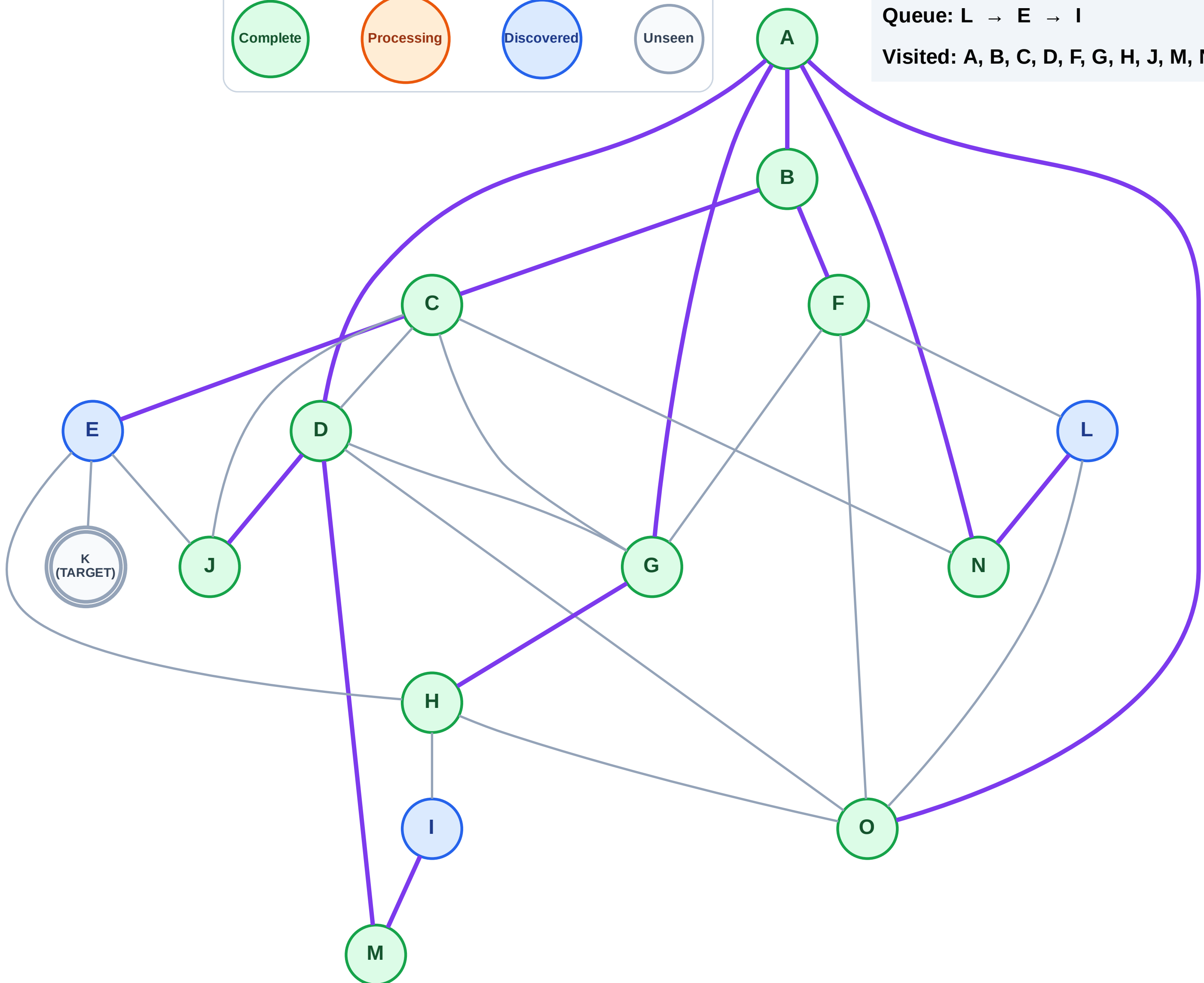
Step 23: No new neighbors from H

Legend



Queue: L → E → I

Visited: A, B, C, D, F, G, H, J, M, N, O



BFS Traversal

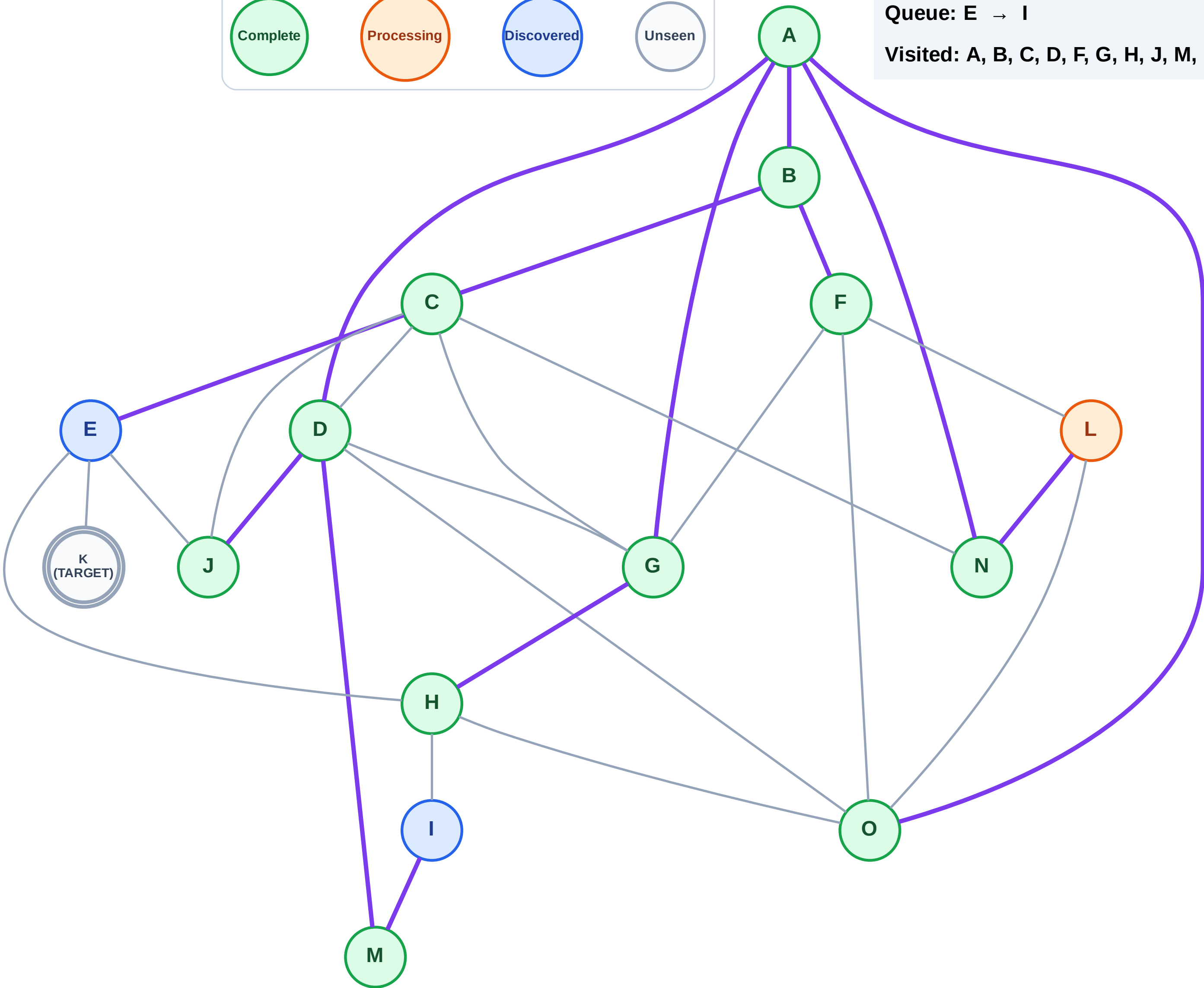
Step 24: Process node L

Legend



Queue: E → I

Visited: A, B, C, D, F, G, H, J, M, N, O



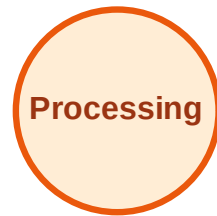
BFS Traversal

Step 25: No new neighbors from L

Legend



Complete



Processing



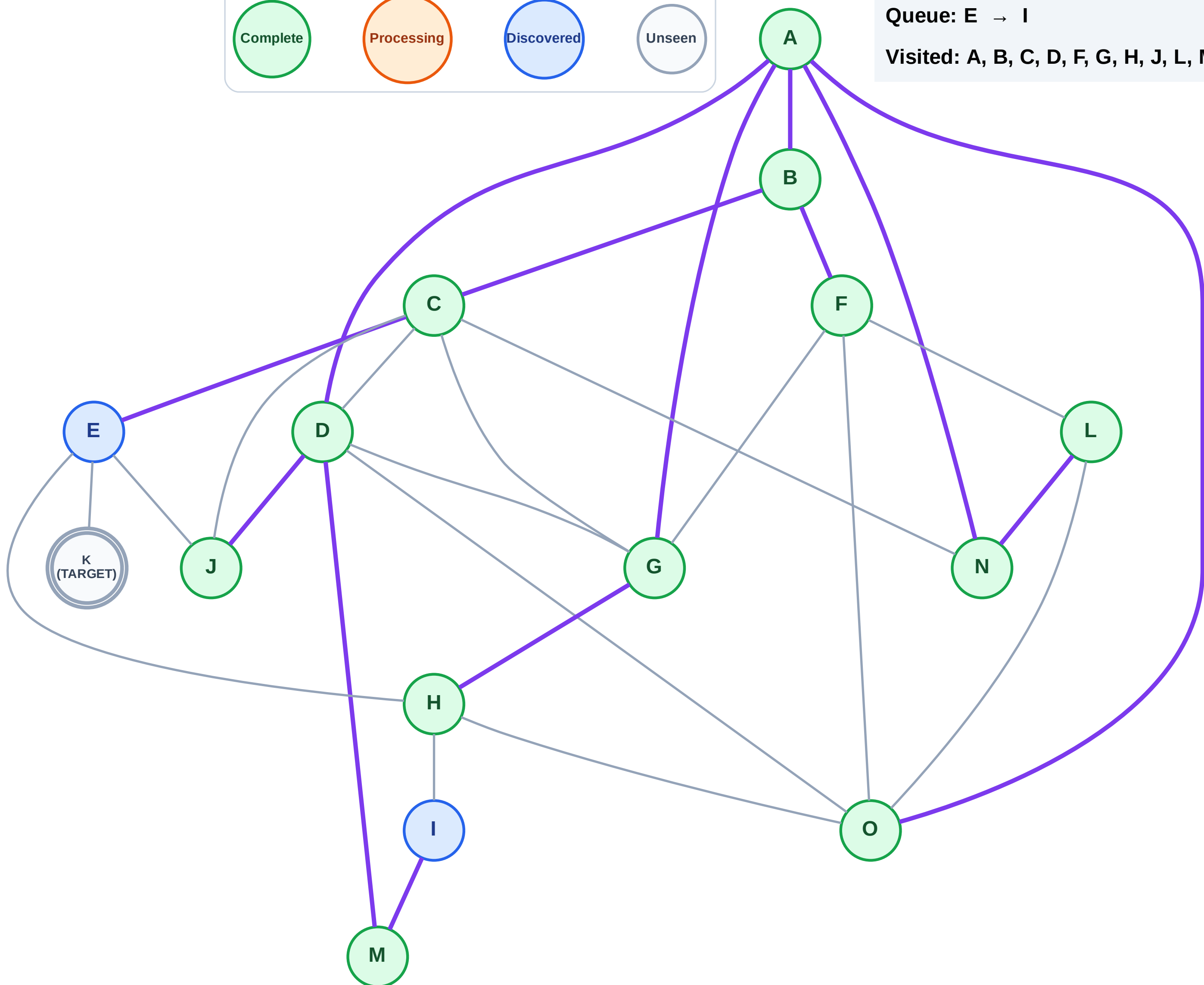
Discovered



Unseen

Queue: E → I

Visited: A, B, C, D, F, G, H, J, L, M, N, O



BFS Traversal

Step 26: Process node E

Legend

Complete

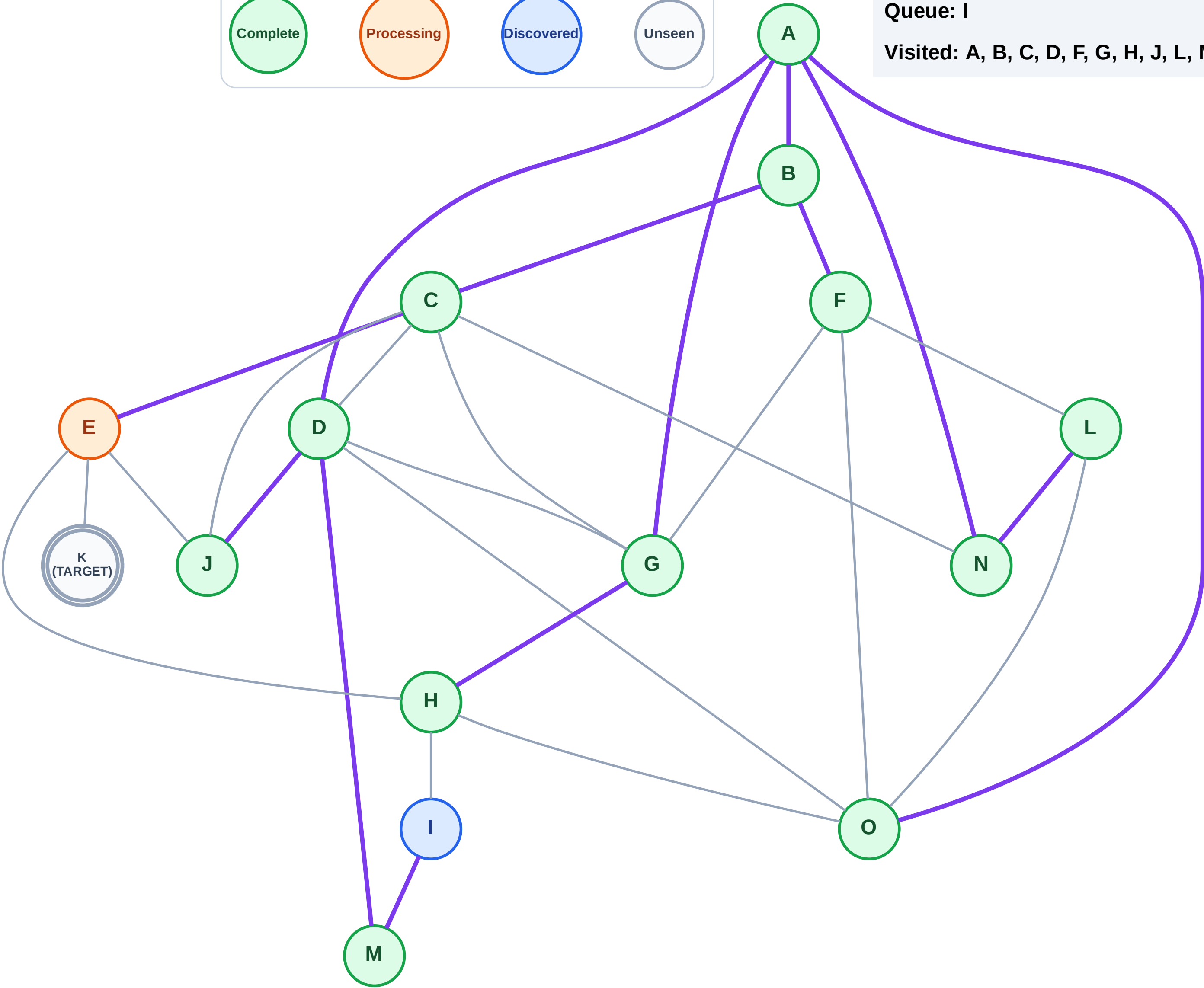
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, F, G, H, J, L, M, N, O



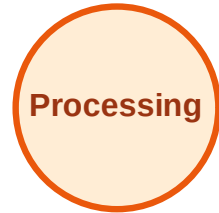
BFS Traversal

Step 27: Discover K from E

Legend



Complete



Processing



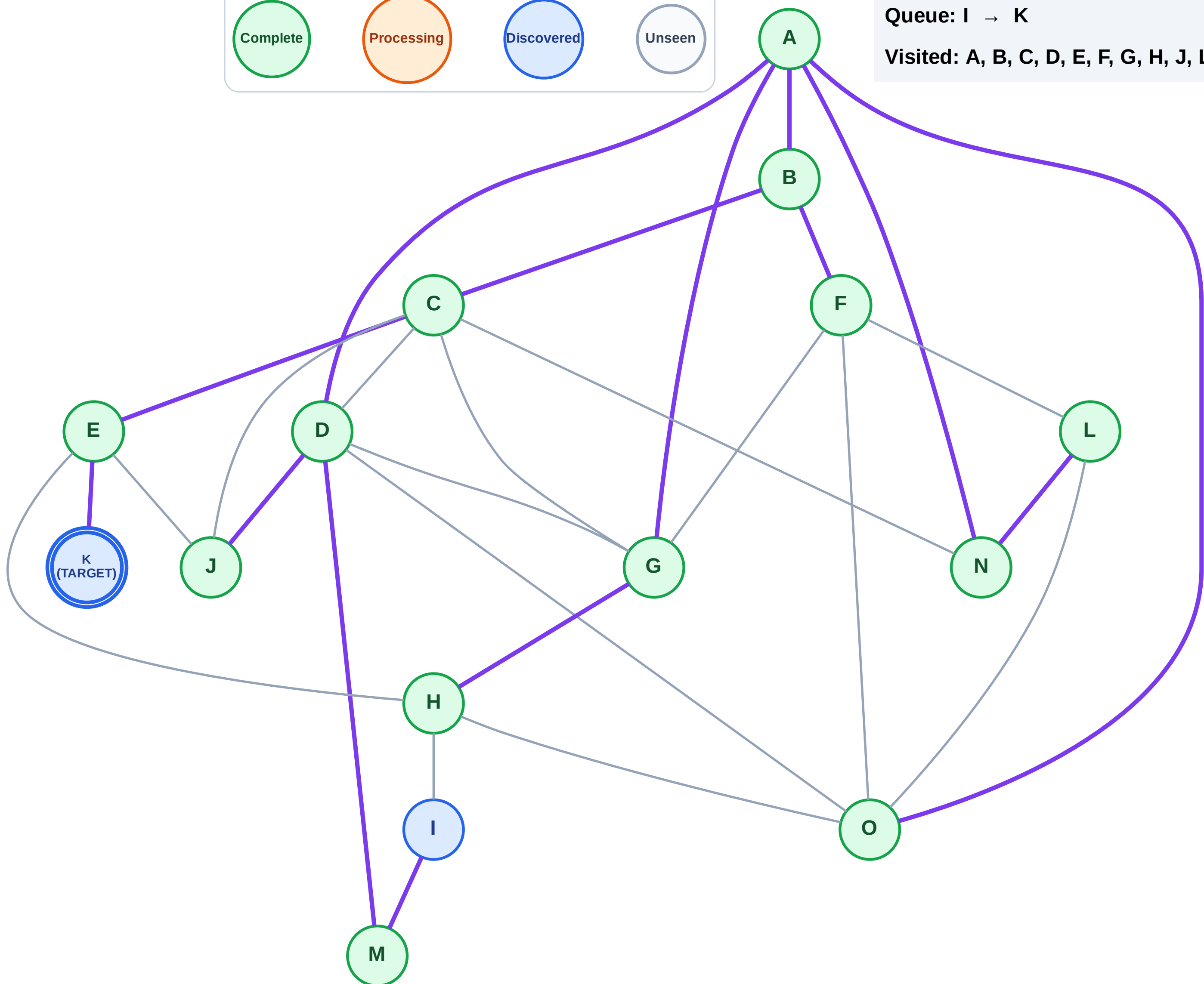
Discovered



Unseen

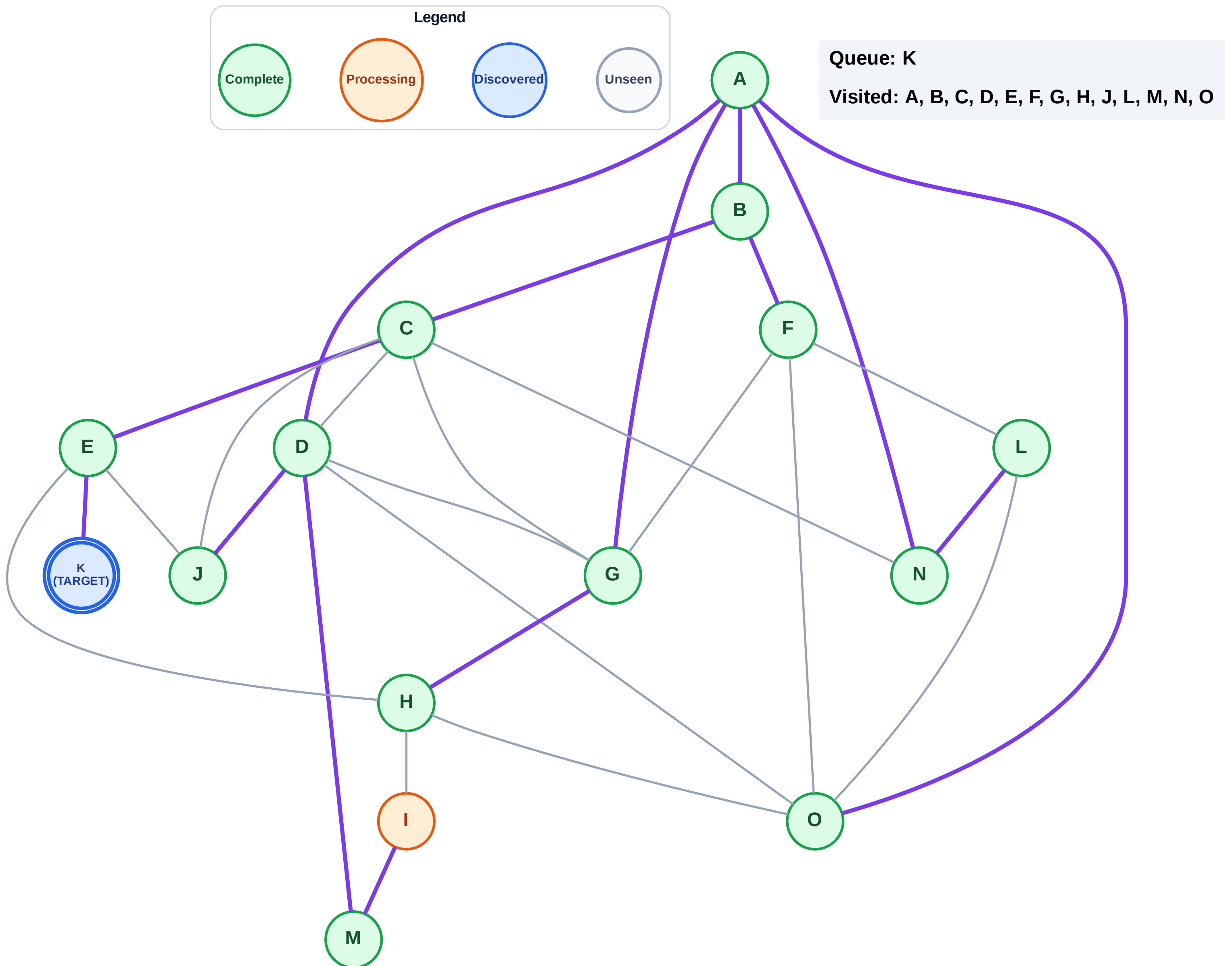
Queue: I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



Step 28: Process node I

Step 28: Process node I



Step 29: No new neighbors from I

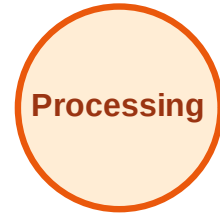
Step 29: No new neighbors from I



BFS Traversal

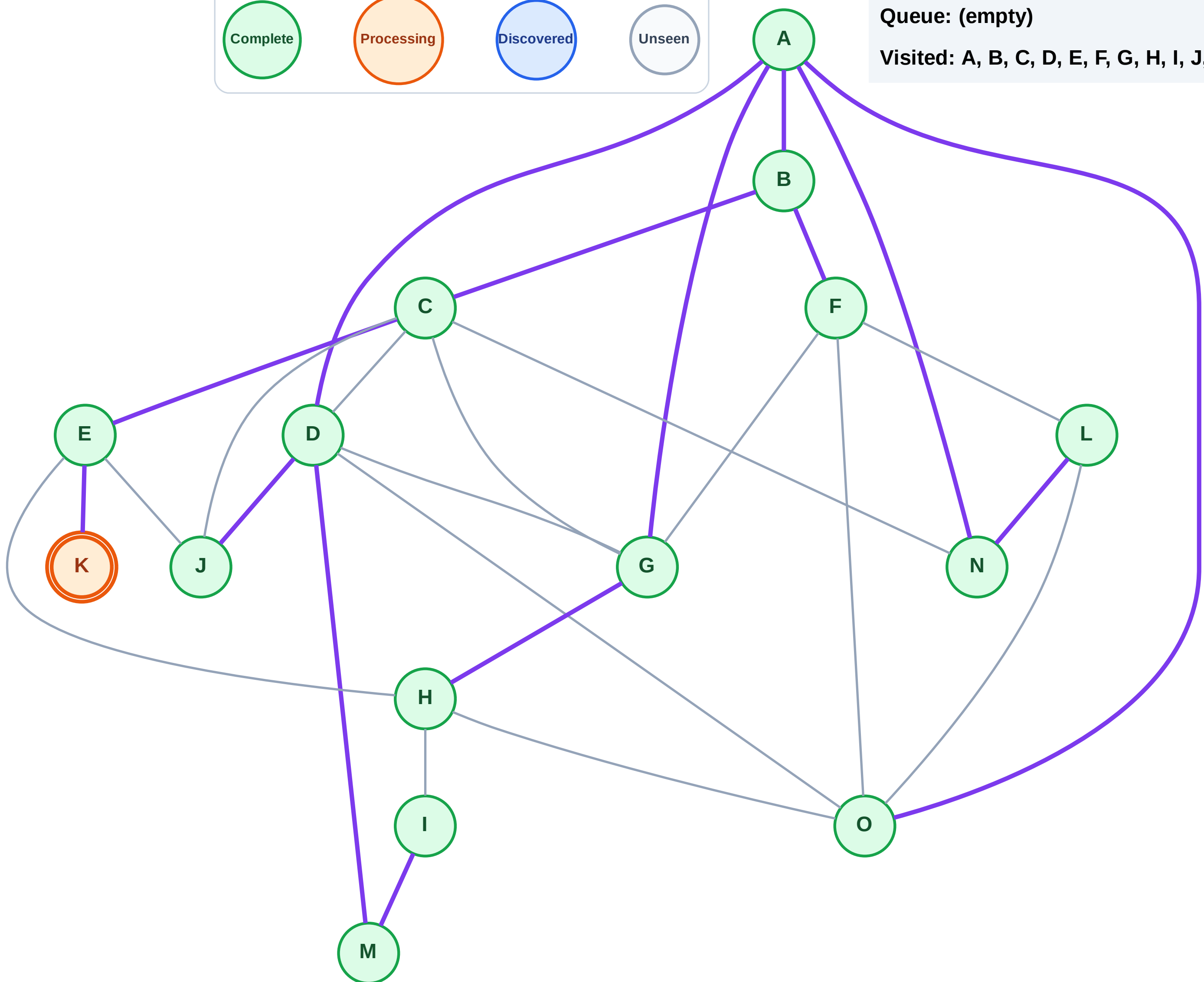
Step 30: Process node K

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (4 hops): A → B → C → E → K

Legend



Complete



Processing



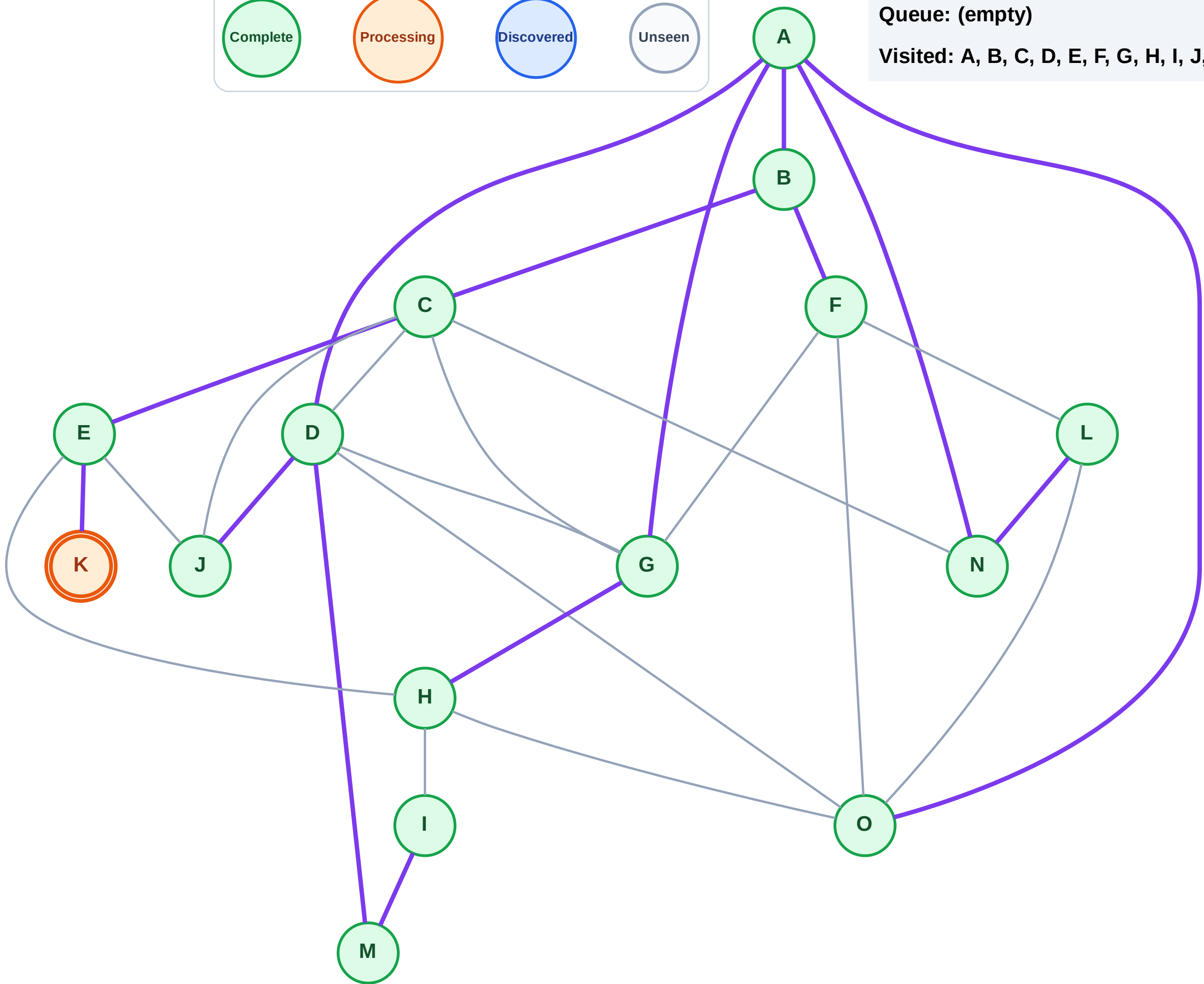
Discovered



Unseen

Queue: (empty)

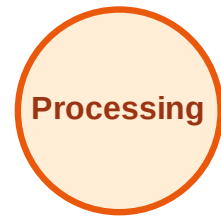
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

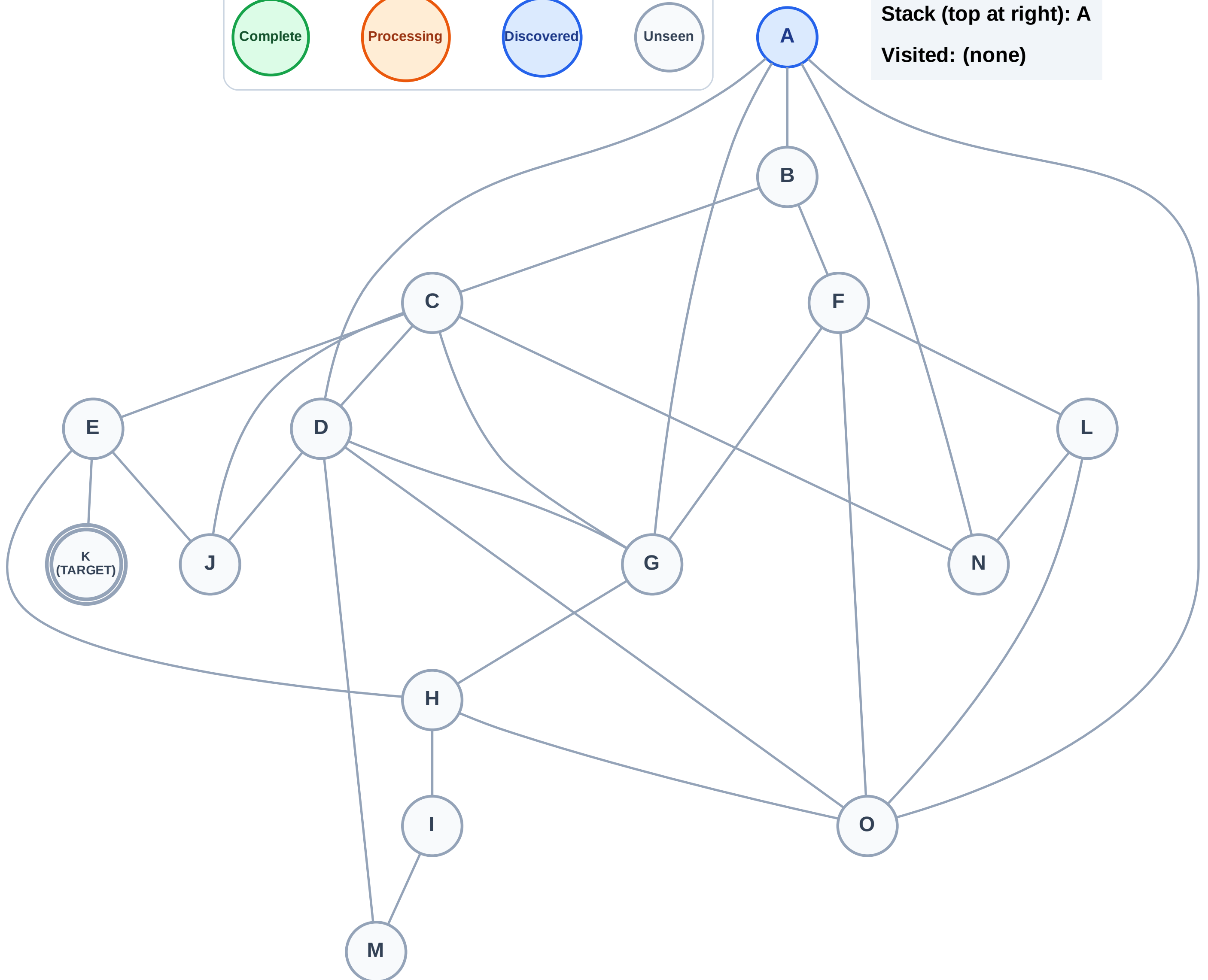
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

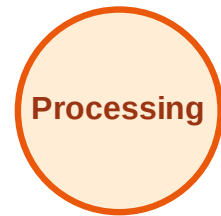
Visited: (none)



DFS Traversal

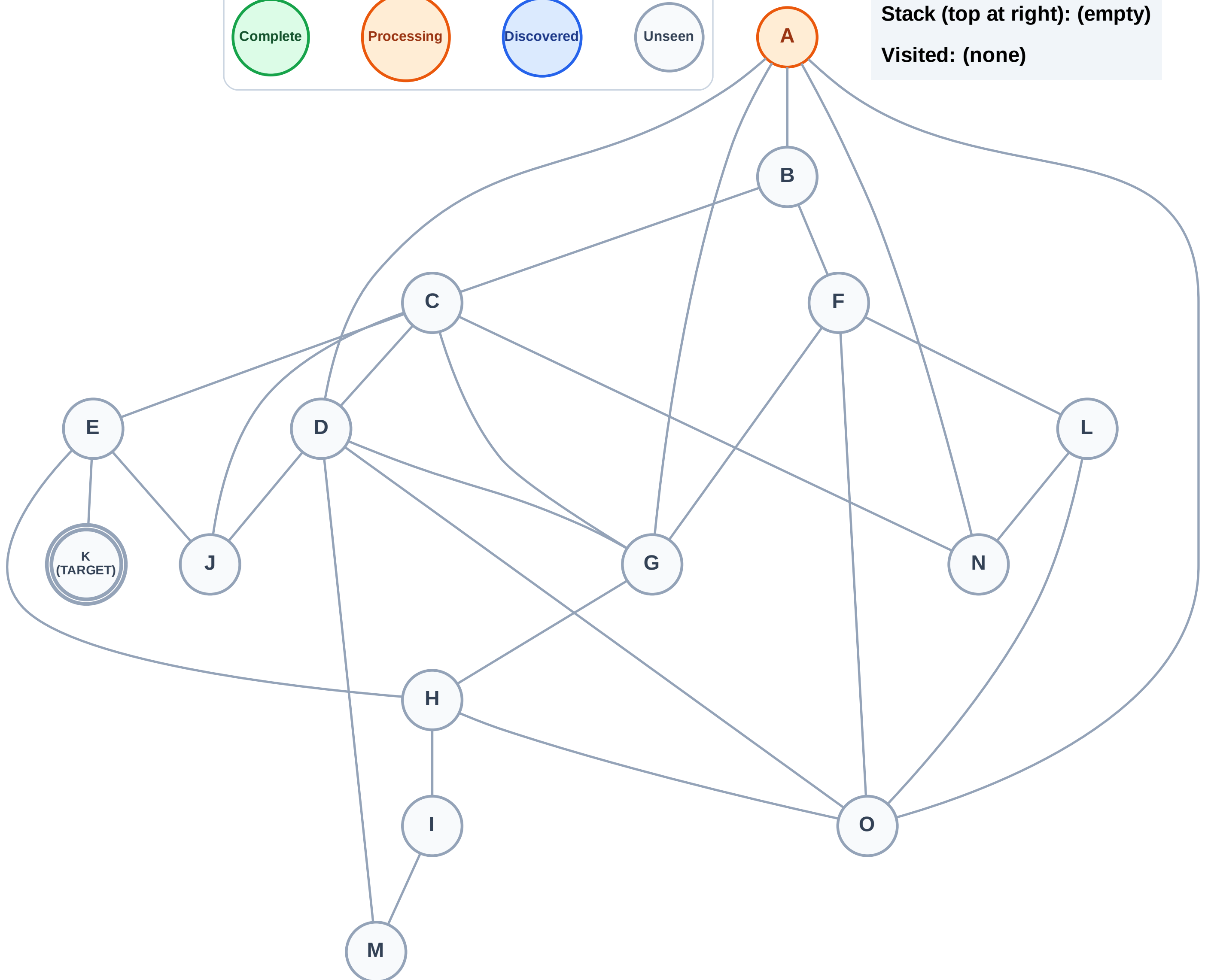
Step 2: Process node A

Legend



Stack (top at right): (empty)

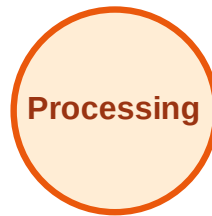
Visited: (none)



DFS Traversal

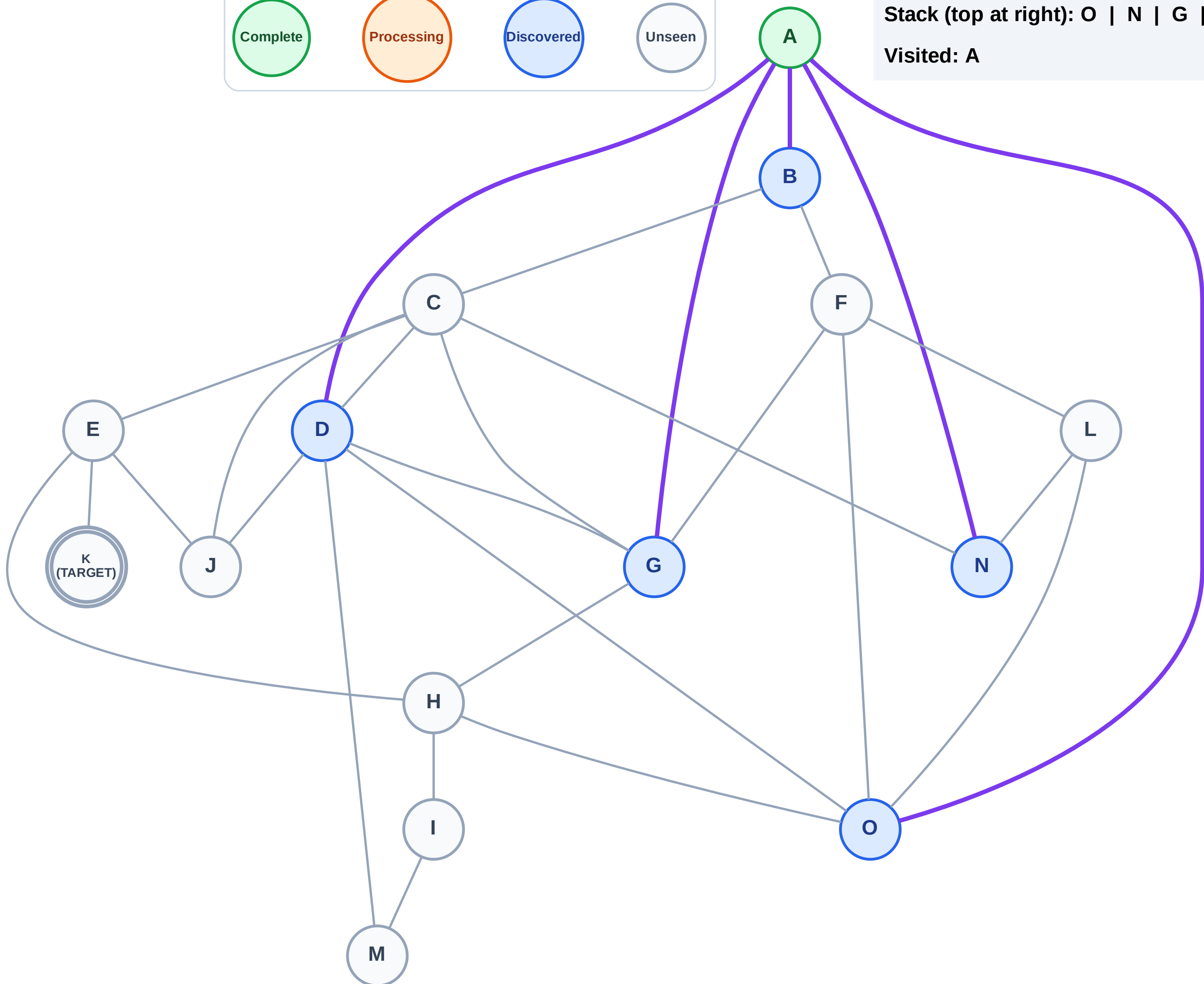
Step 3: Discover O, N, G, D, B from A

Legend



Stack (top at right): O | N | G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

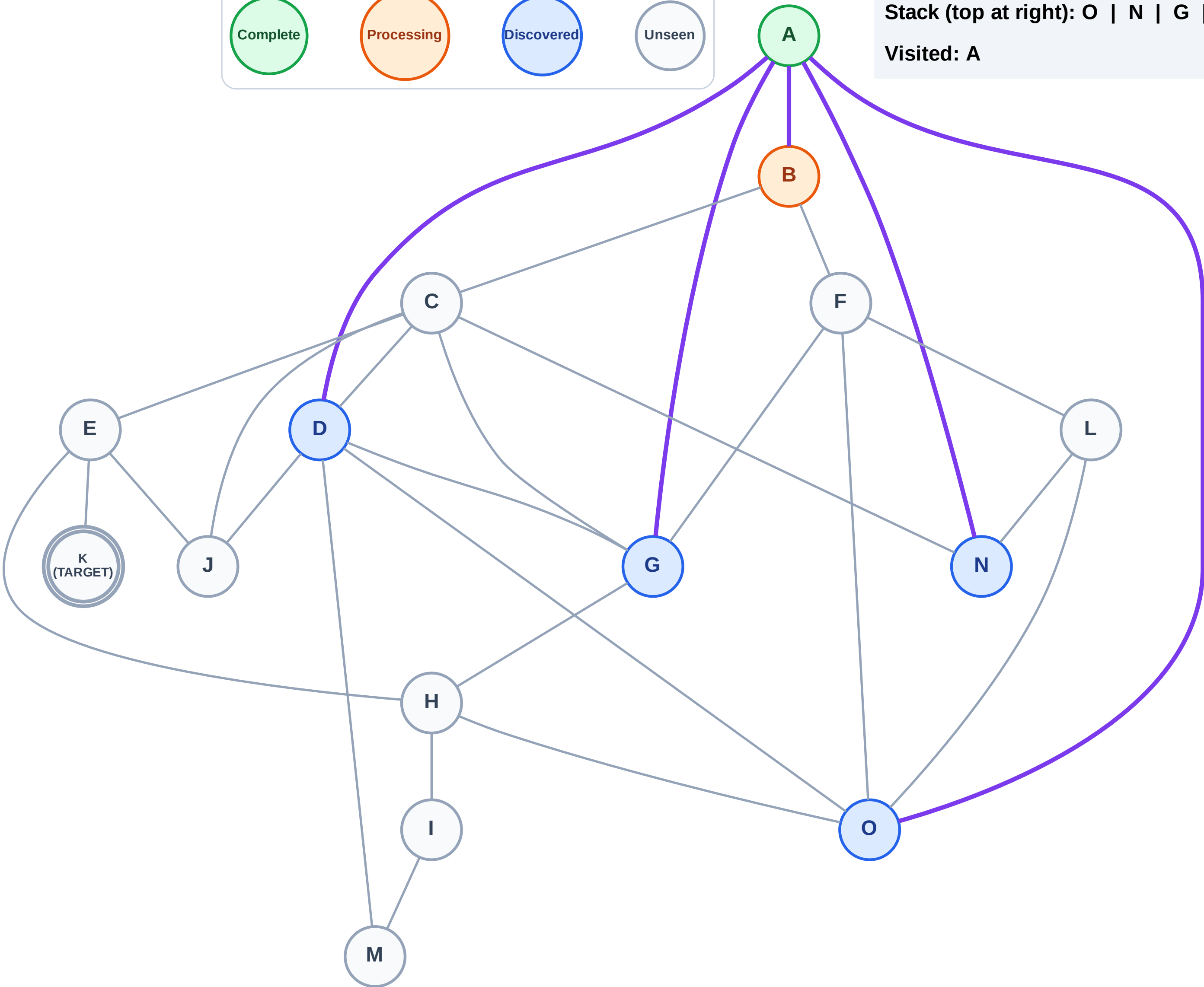
Processing

Discovered

Unseen

Stack (top at right): O | N | G | D

Visited: A



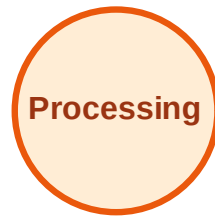
DFS Traversal

Step 5: Discover F, C from B

Legend



Complete



Processing



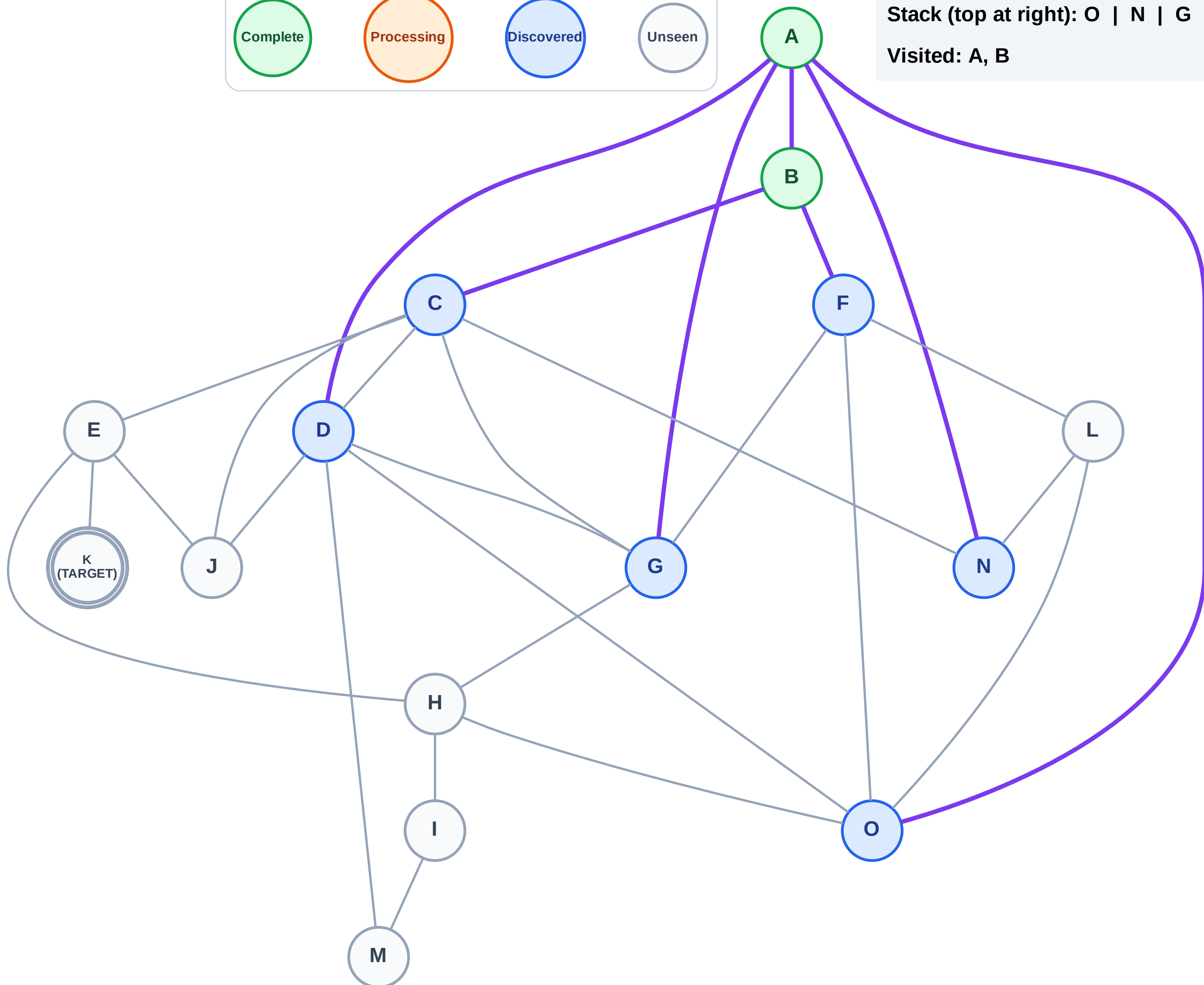
Discovered



Unseen

Stack (top at right): O | N | G | D | F | C

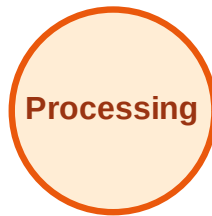
Visited: A, B



DFS Traversal

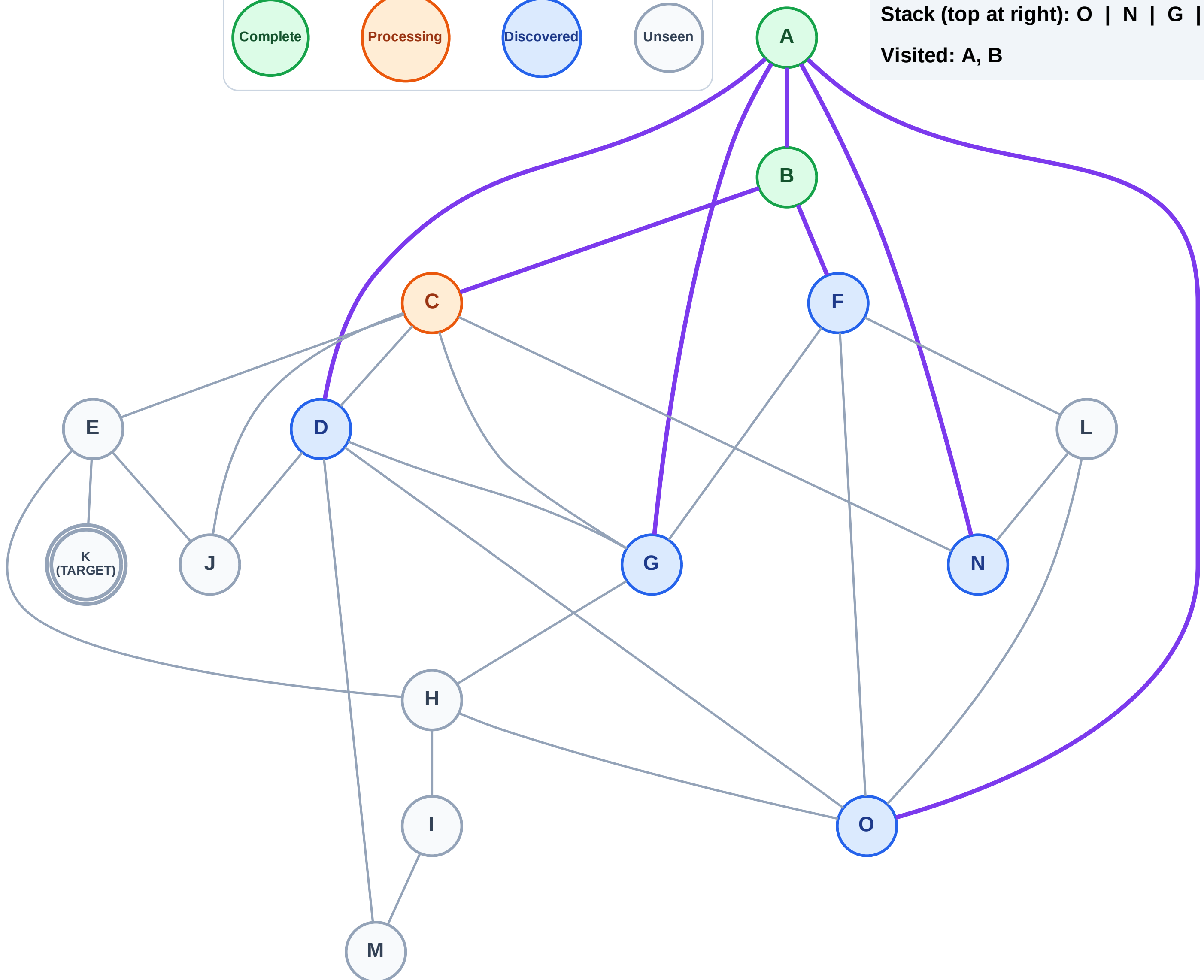
Step 6: Process node C

Legend



Stack (top at right): O | N | G | D | F

Visited: A, B



Step 7: Discover J, E from C

Step 7: Discover J, E from C

Complete

Complete



Processing

Processing



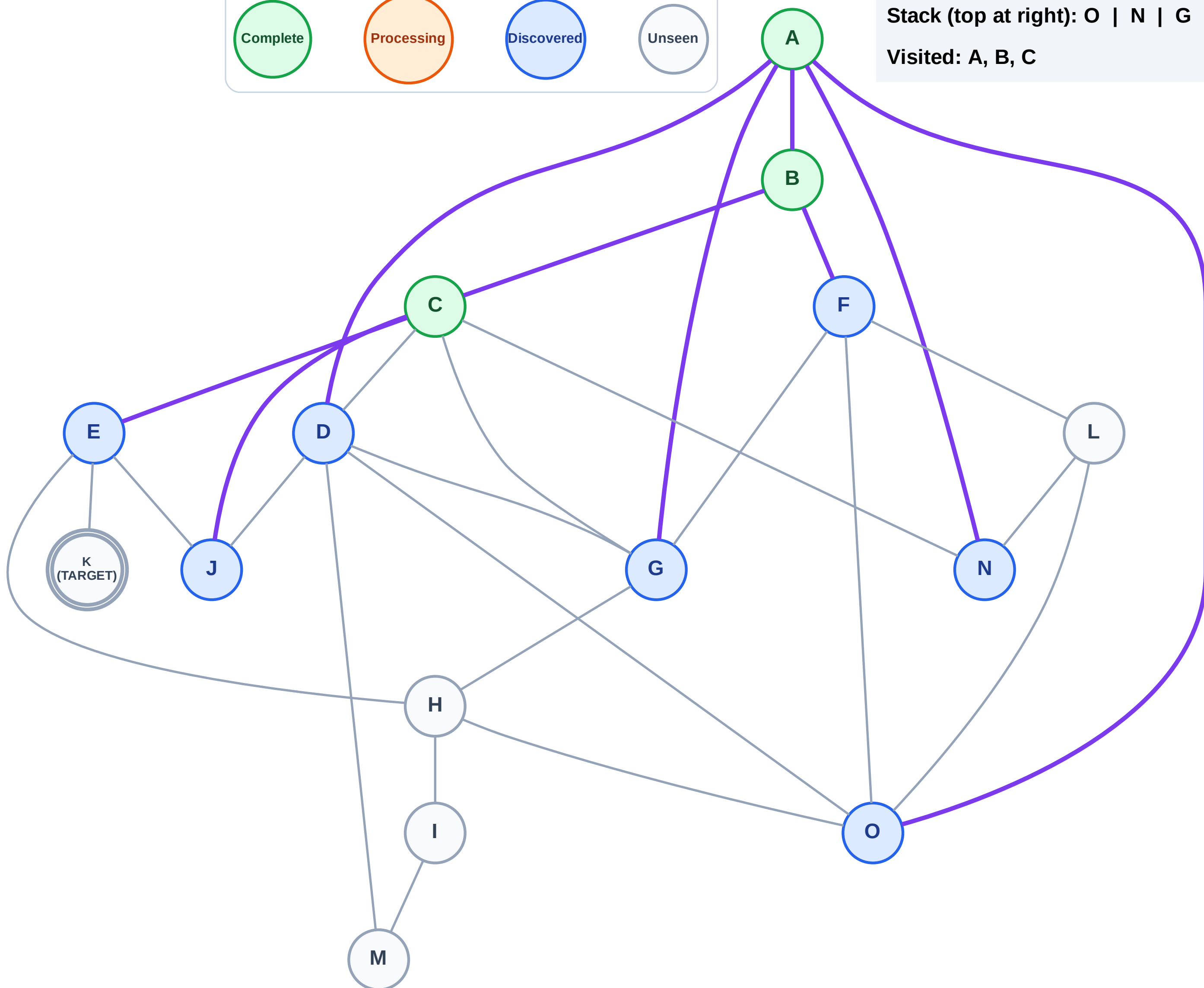
Discovered

Discovered



Unseen

Visited: A, B, C



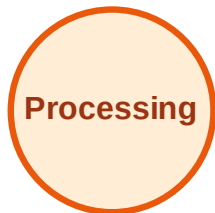
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



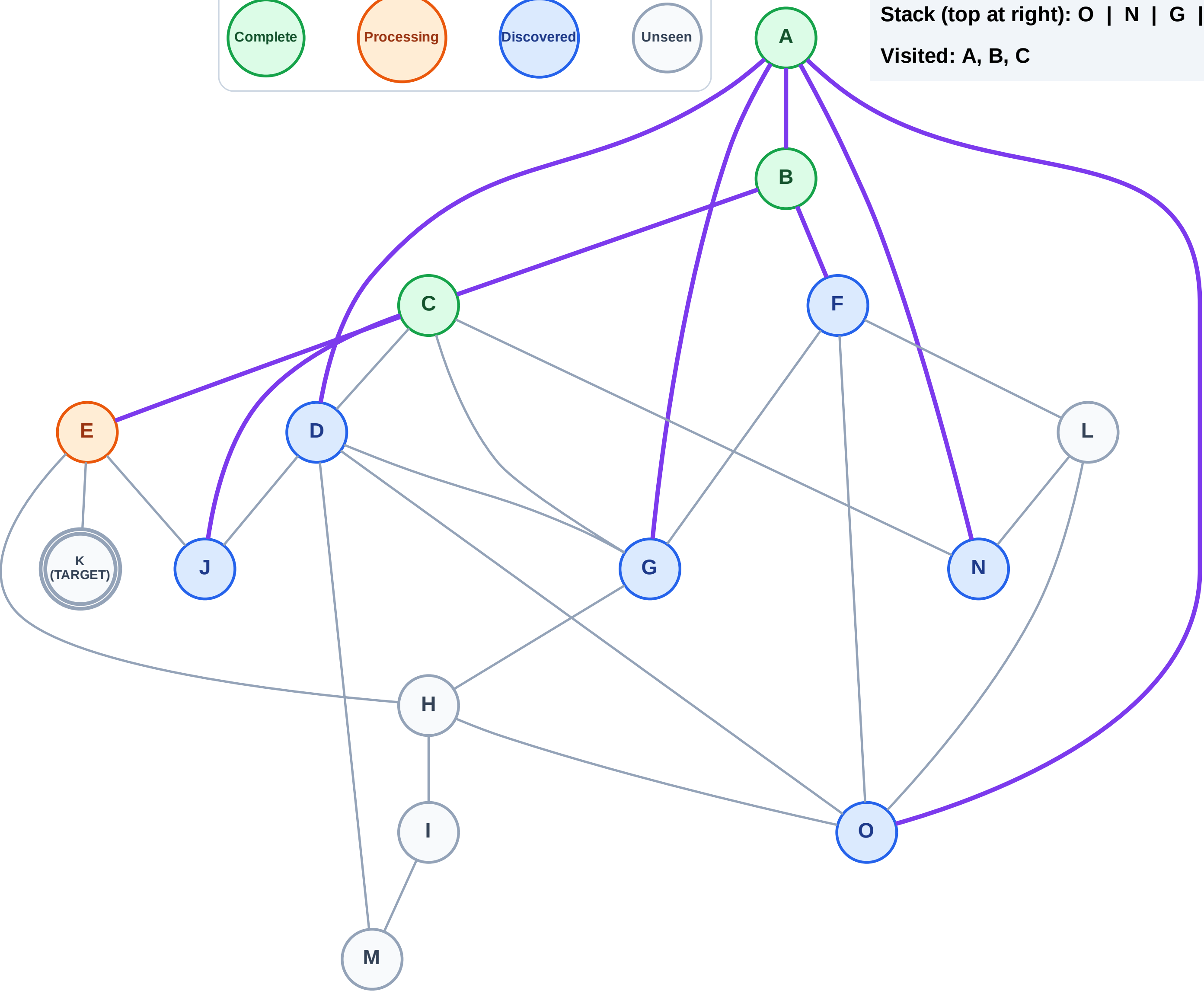
Discovered



Unseen

Stack (top at right): O | N | G | D | F | J

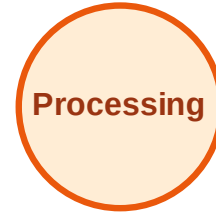
Visited: A, B, C



DFS Traversal

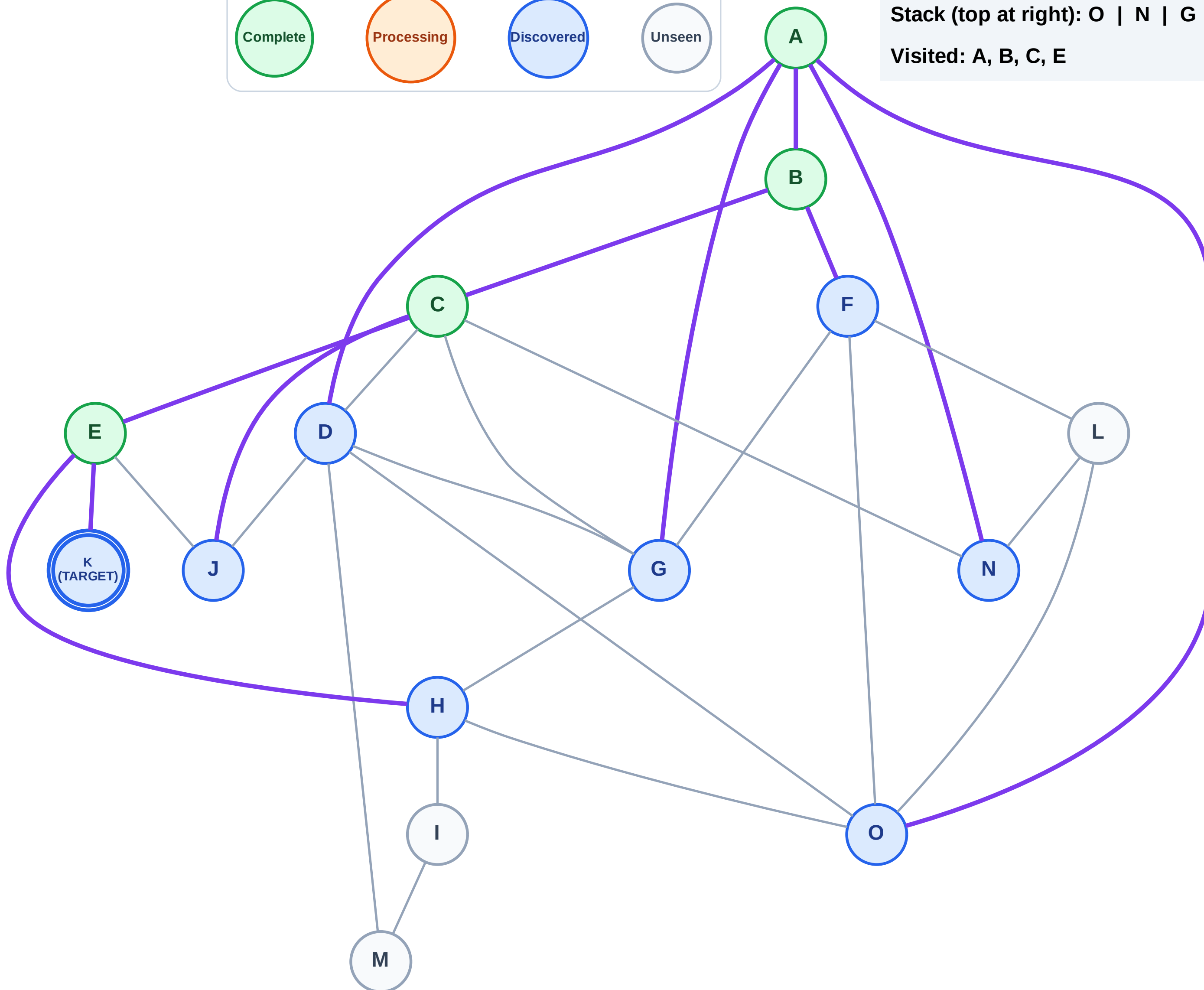
Step 9: Discover K, H from E

Legend



Stack (top at right): O | N | G | D | F | J | K | H

Visited: A, B, C, E



DFS Traversal

Step 10: Process node H

Legend

Complete

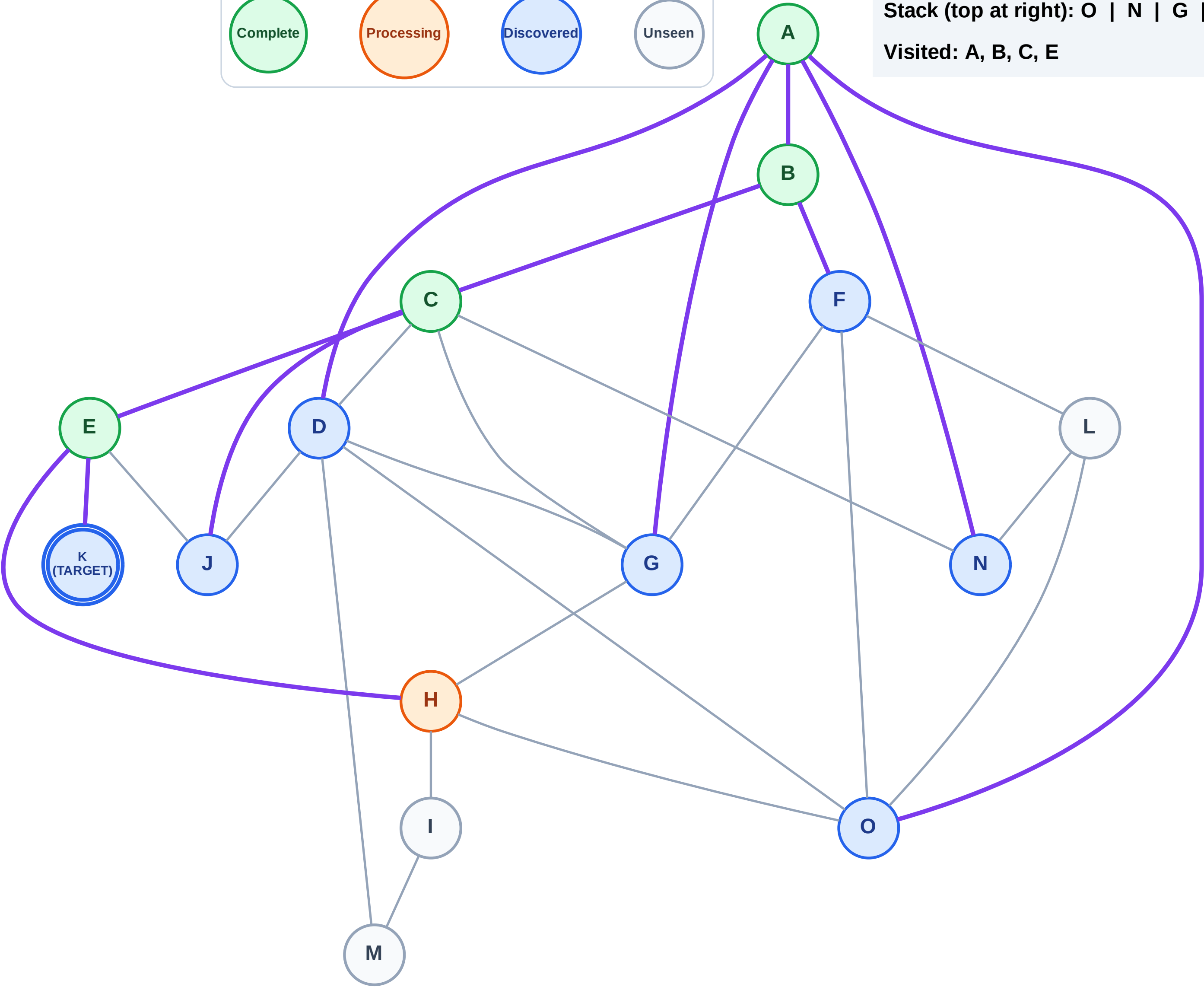
Processing

Discovered

Unseen

Stack (top at right): O | N | G | D | F | J | K

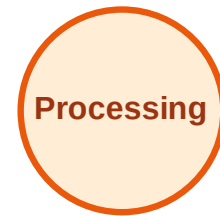
Visited: A, B, C, E



DFS Traversal

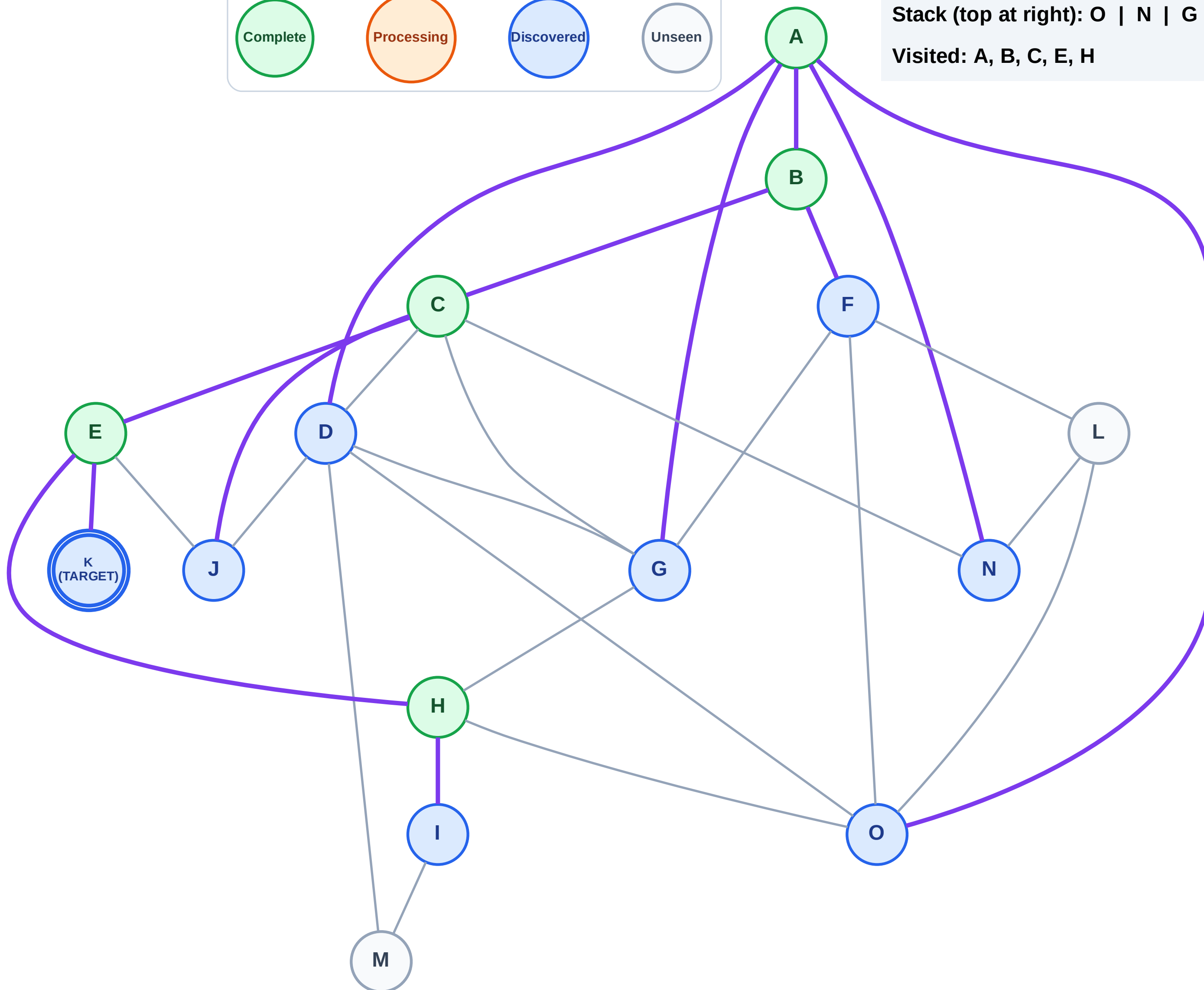
Step 11: Discover I from H

Legend



Stack (top at right): O | N | G | D | F | J | K | I

Visited: A, B, C, E, H



DFS Traversal

Step 12: Process node I

Legend

Complete

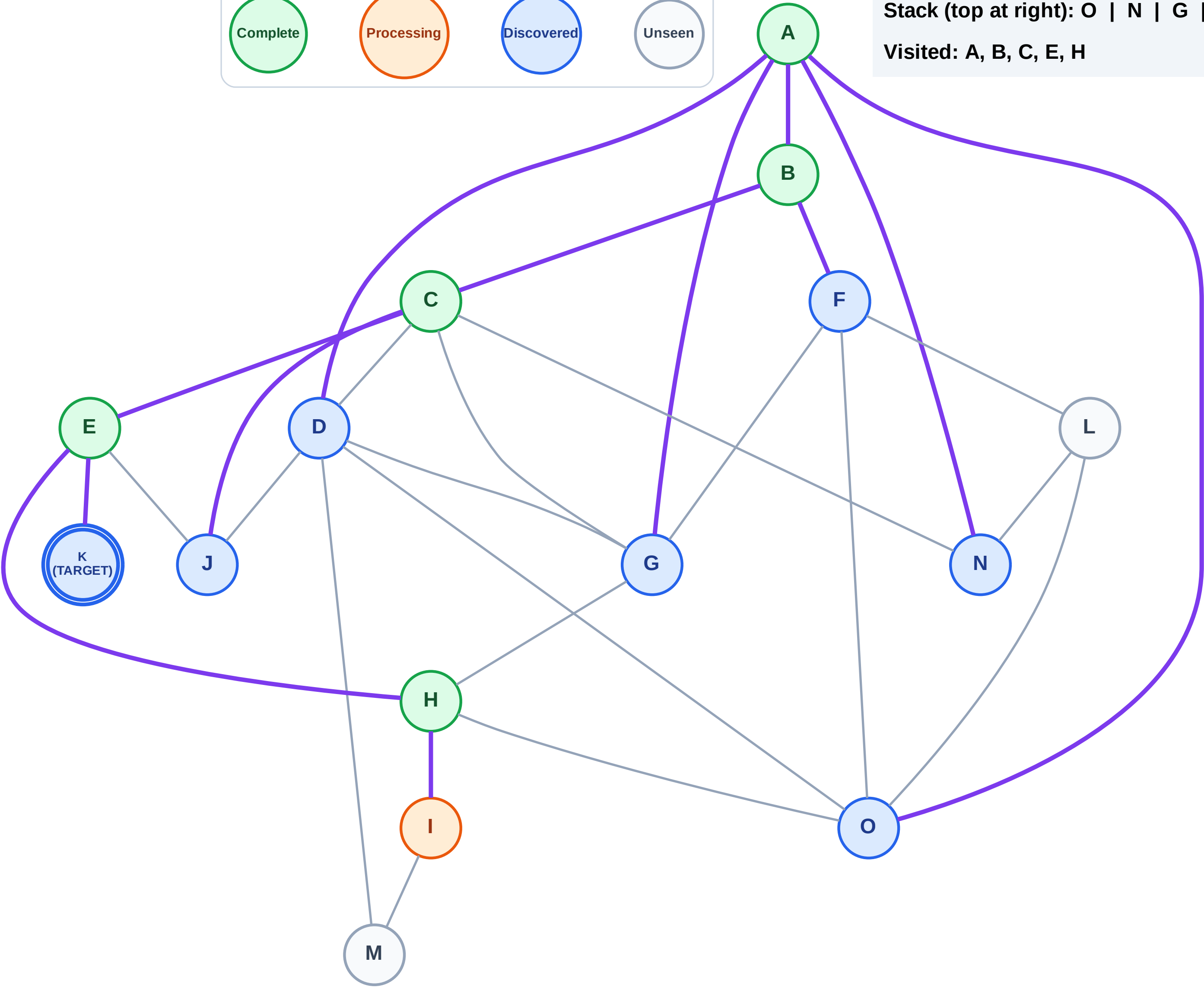
Processing

Discovered

Unseen

Stack (top at right): O | N | G | D | F | J | K

Visited: A, B, C, E, H



DFS Traversal

Step 13: Discover M from I

Legend

Complete



Processing

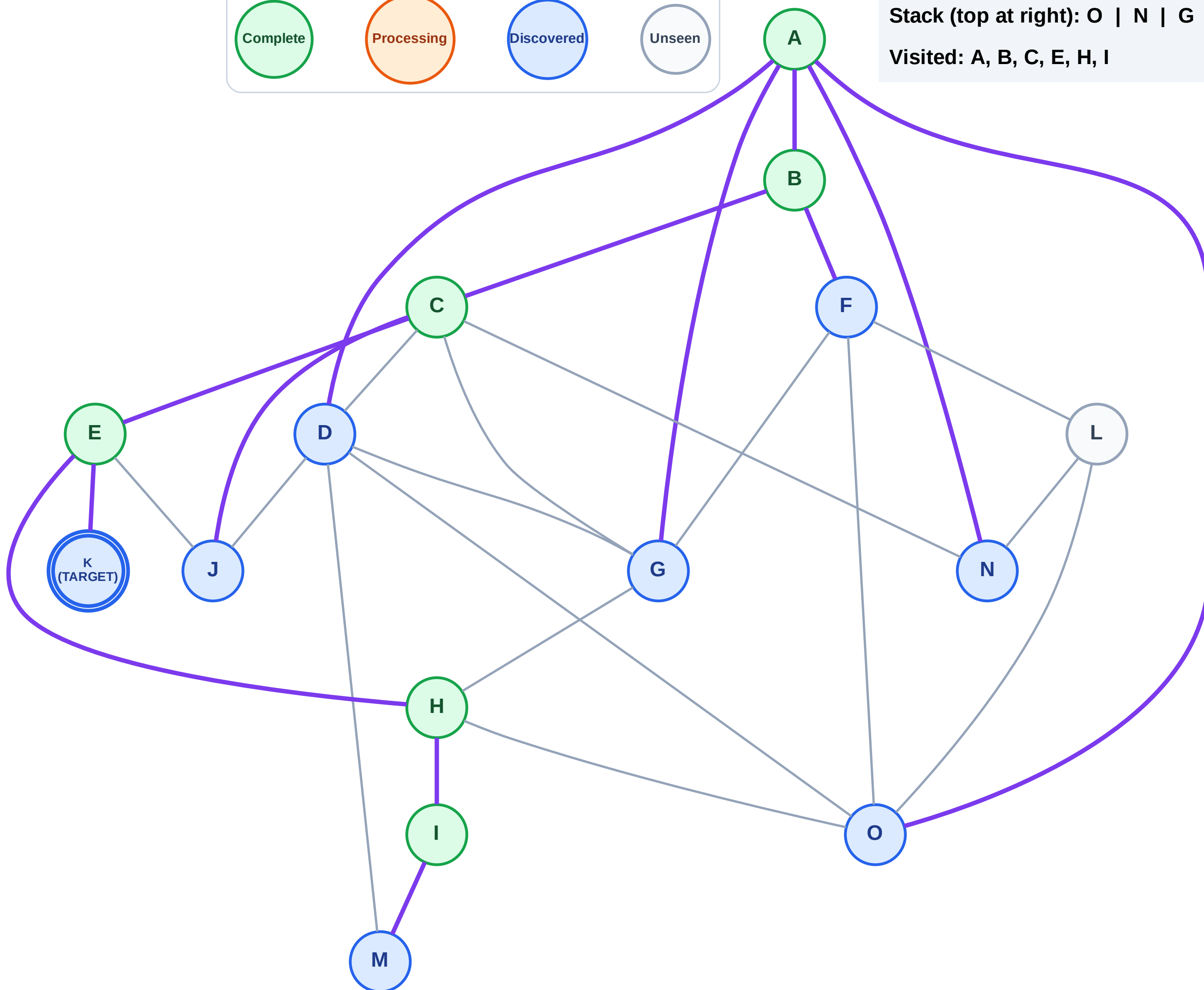


Discovered

Unseen

Stack (top at right): O | N | G | D | F | J | K | M

Visited: A, B, C, E, H, I



DFS Traversal

Step 14: Process node M

Legend

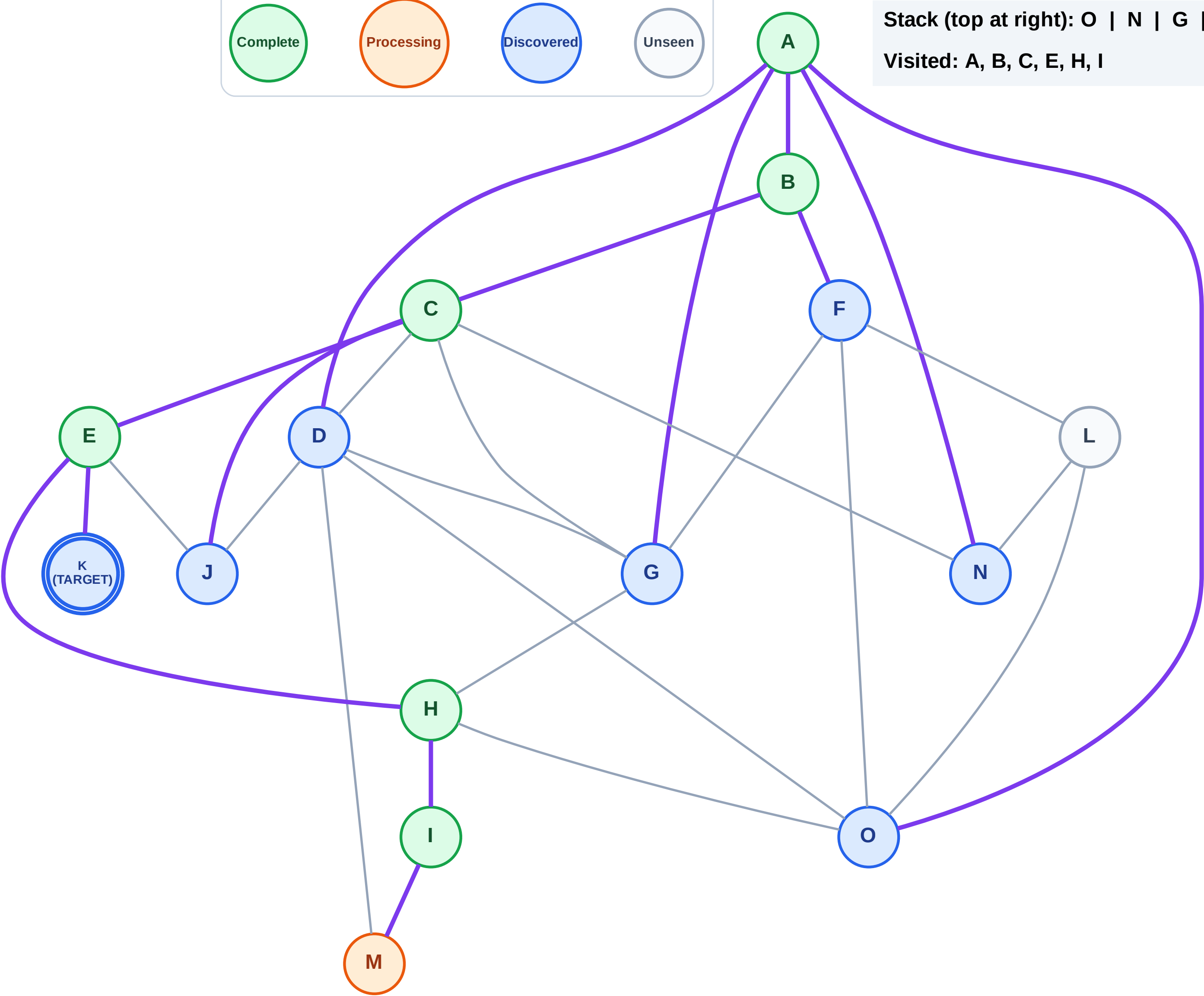
Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | G | D | F | J | K
Visited: A, B, C, E, H, I



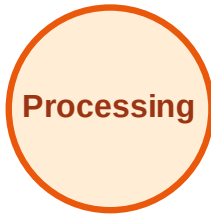
DFS Traversal

Step 15: No new neighbors from M

Legend



Complete



Processing



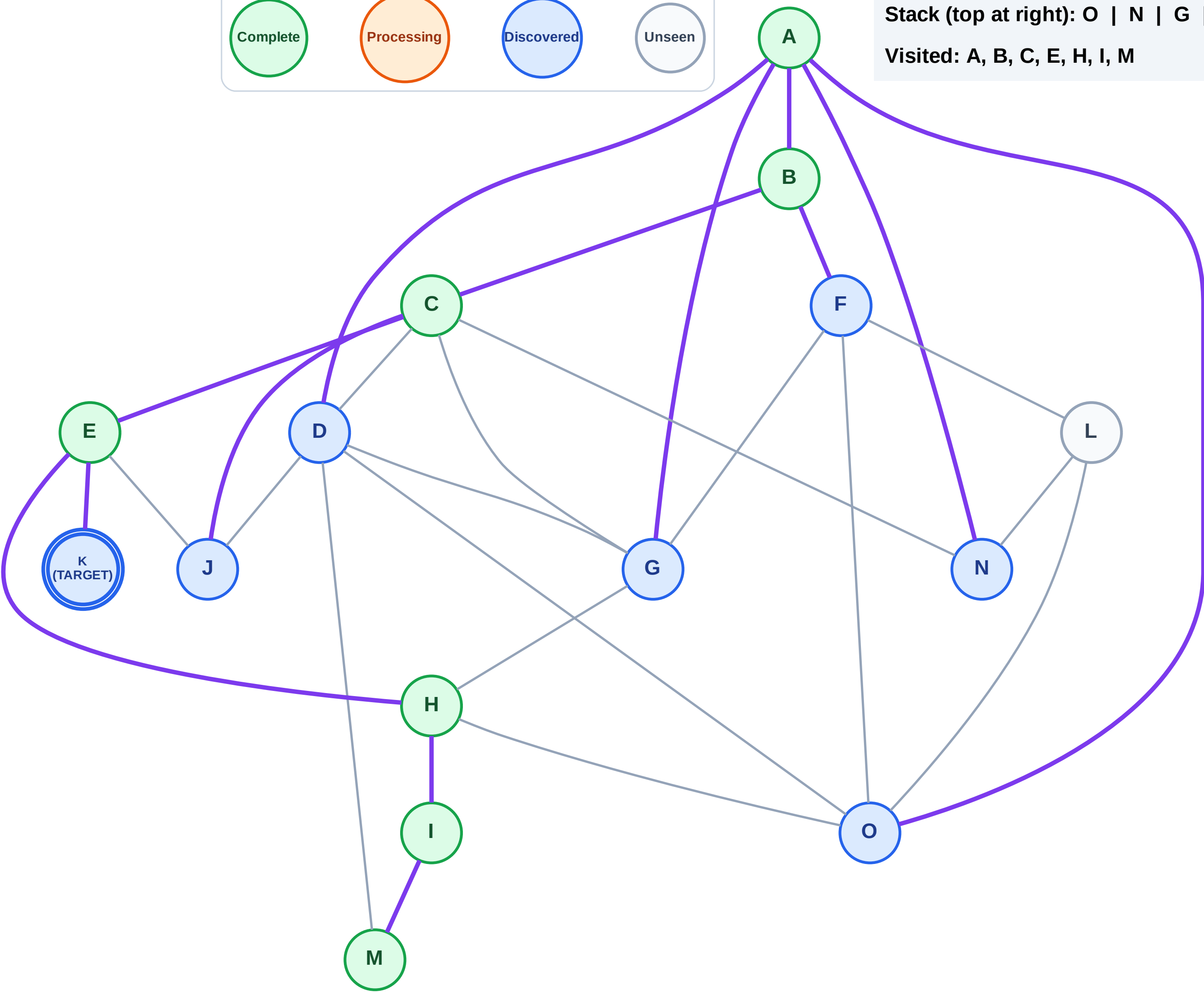
Discovered



Unseen

Stack (top at right): O | N | G | D | F | J | K

Visited: A, B, C, E, H, I, M



Step 16: Process node K



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → C → E → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): O | N | G | D | F | J

Visited: A, B, C, E, H, I, K, M

